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EasyVisa

Machine Learning II

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Contents

Context: 4
Objective: 4
Data Description 4
Understand the shape of the dataset 5
Check the data types of the columns for the dataset 5
Exploratory Data Analysis 5
Univariate Analysis 7
Observations on number of employees 7
Observations on prevailing wage 8
Observations on continent 8
Observations on education of employee 9
Observations on job experience 10
Observations on job training 10
Observations on region of employment 11
Observations on unit of wage 11
Observations on case status 12
Bivariate Analysis 12
Outlier Check 19
Data Preparation for modeling 19
Model evaluation criterion 20
Model can make wrong predictions as: 20
Which case is more important? 20
How to reduce the losses? 20
Decision Tree Model 21
Hyperparameter Tuning - Decision Tree performance 22
Bagging Classifier 24
Hyperparameter Tuning - Bagging Classifier 25
Random Forest 27
Hyperparameter Tuning - Random Forest 30
AdaBoost Classifier 32
Hyperparameter Tuning - AdaBoost Classifier 34
Gradient Boosting Classifier 36
Hyperparameter Tuning - Gradient Boosting Classifier 38

XGBoost Classifier 40
Hyperparameter Tuning - XGBoost Classifier 42
Stacking Classifier 43
Conclusion 45
Important features of the final model 47
Recommendations 47

Table of figures

Figure 1(no. of employees) 7
Figure 2(prevailing wage) 8
Figure 3(continents) 8
Figure 4(education of employees) 9
Figure 5(job experience) 10
Figure 6(require job training) 10
Figure 7(region of employment) 11
Figure 8(unit of wage) 11
Figure 9(case status) 12
Figure 10(confusion matrix) 12
Figure 11(edu vs visa) 13
Figure 12(region vs edu) 14
Figure 13(region vs visa) 14
Figure 14(continent vs visa) 15
Figure 15(work ex vs visa) 16
Figure 16(work ex vs training) 16
Figure 17(wage vs visa) 17
Figure 18(wage vs region) 18
Figure 19(unit of wage vs visa) 19
Figure 20(outliers check) 19
Figure 21(dec. tree performance train) 21
Figure 22(dec. tree performance test) 21
Figure 23(dec. tree performance tuned train) 22
Figure 24(dec. tree performance tuned test) 23
Figure 25(bagging cls train) 24
Figure 26(bagging cls test) 25
Figure 27(bagging cls tuned train) 26
Figure 28(bagging cls tuned test) 27
Figure 29(Random forest train) 28
Figure 30(Random forest test) 29
Figure 31(Random forest tuned train) 30
Figure 32(Random forest tuned test) 31
Figure 33(adaboost train) 32
Figure 34((adaboost test) 33
Figure 35(adaboost tuned train) 34

Figure 36(adaboost tuned test) 35
Figure 37(gradient boost train) 36
Figure 38(gradient boost test) 37
Figure 39(gradient boost tuned train) 38
Figure 40(gradient boost tuned test) 39
Figure 41(xgboost train) 40
Figure 42(gradient boost test) 41
Figure 43(gradient boost tuned train) 42
Figure 44(gradient boost tuned test) 43
Figure 45(stacking cls train) 44
Figure 46(stacking cls test) 45
Figure 47(all model performance) 46
Figure 48(feature importance) 47

Context:

Business communities in the United States are facing high demand for human resources, but one of the constant challenges is identifying and attracting the right talent, which is perhaps the most important element in remaining competitive. Companies in the United States look for hard-working, talented, and qualified individuals both locally as well as abroad.

The Immigration and Nationality Act (INA) of the US permits foreign workers to come to the United States to work on either a temporary or permanent basis. The act also protects US workers against adverse impacts on their wages or working conditions by ensuring US employers' compliance with statutory requirements when they hire foreign workers to fill workforce shortages. The immigration programs are administered by the Office of Foreign Labor Certification (OFLC).

OFLC processes job certification applications for employers seeking to bring foreign workers into the United States and grants certifications in those cases where employers can demonstrate that there are not sufficient US workers available to perform the work at wages that meet or exceed the wage paid for the occupation in the area of intended employment.

Objective:

In FY 2016, the OFLC processed 775,979 employer applications for 1,699,957 positions for temporary and permanent labor certifications. This was a nine percent increase in the overall number of processed applications from the previous year. The process of reviewing every case is becoming a tedious task as the number of applicants is increasing every year.

The increasing number of applicants every year calls for a Machine Learning based solution that can help in shortlisting the candidates having higher chances of VISA approval. OFLC has hired the firm EasyVisa for data-driven solutions. You as a data scientist at EasyVisa have to analyze the data provided and, with the help of a classification model:

- Facilitate the process of visa approvals.
- Recommend a suitable profile for the applicants for whom the visa should be certified or denied based on the drivers that significantly influence the case status.

Data Description

The data contains the different attributes of employee and the employer. The detailed data dictionary is given below.

- case_id: ID of each visa application
- continent: Information of continent the employee
- education_of_employee: Information of education of the employee
- has_job_experience: Does the employee has any job experience? Y= Yes; N = No
- requires_job_training: Does the employee require any job training? Y = Yes; N = No
- no_of_employees: Number of employees in the employer's company

- yr_of_estab: Year in which the employer's company was established
- region_of_employment: Information of foreign worker's intended region of employment in the US.
- prevailing_wage: Average wage paid to similarly employed workers in a specific occupation in the area of intended employment. The purpose of the prevailing wage is to ensure that the foreign worker is not underpaid compared to other workers offering the same or similar service in the same area of employment.
- unit_of_wage: Unit of prevailing wage. Values include Hourly, Weekly, Monthly, and Yearly.
- full_time_position: Is the position of work full-time? Y = Full Time Position; N = Part Time Position
- case_status: Flag indicating if the Visa was certified or denied

Understand the shape of the dataset

```
(25480, 12)
```

Check the data types of the columns for the dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25480 entries, 0 to 25479
Data columns (total 12 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   case_id                25480 non-null  object
 1   continent              25480 non-null  object
 2   education_of_employee  25480 non-null  object
 3   has_job_experience     25480 non-null  object
 4   requires_job_training  25480 non-null  object
 5   no_of_employees        25480 non-null  int64
 6   yr_of_estab            25480 non-null  int64
 7   region_of_employment   25480 non-null  object
 8   prevailing_wage        25480 non-null  float64
 9   unit_of_wage           25480 non-null  object
 10  full_time_position     25480 non-null  object
 11  case_status            25480 non-null  object
dtypes: float64(1), int64(2), object(9)
memory usage: 2.3+ MB
```

```
# checking for duplicate values

np.int64(0)
```

Exploratory Data Analysis

Let's check the statistical summary of the data.

```
data.describe()
```

Fixing the negative values in number of employees columns

Number of rows fixed is 33

```
(33, 12)
```

Let's check the count of each unique category in each of the categorical variables.

```
case_id
EZYV25480    1
EZYV01       1
EZYV02       1
EZYV03       1
EZYV04       1
            ..
EZYV13       1
EZYV12       1
EZYV11       1
EZYV10       1
EZYV09       1
Name: count, Length: 25480, dtype: int64
--------------------------------------------------
continent
Asia             16861
Europe            3732
North America     3292
South America      852
Africa             551
Oceania            192
Name: count, dtype: int64
--------------------------------------------------
education_of_employee
Bachelor's     10234
Master's        9634
High School     3420
Doctorate       2192
Name: count, dtype: int64
--------------------------------------------------
has_job_experience
Y    14802
N    10678
Name: count, dtype: int64
--------------------------------------------------
requires_job_training
N    22525
Y     2955
Name: count, dtype: int64
--------------------------------------------------
region_of_employment
Northeast    7195
South        7017
```

```
West         6586
Midwest      4307
Island        375
Name: count, dtype: int64
--------------------------------------------------
unit_of_wage
Year     22962
Hour      2157
Week       272
Month       89
Name: count, dtype: int64
--------------------------------------------------
full_time_position
Y    22773
N     2707
Name: count, dtype: int64
--------------------------------------------------
case_status
Certified    17018
Denied        8462
Name: count, dtype: int64
--------------------------------------------------
```

```
# dropping case_id
```

Univariate Analysis

Observations on number of employees

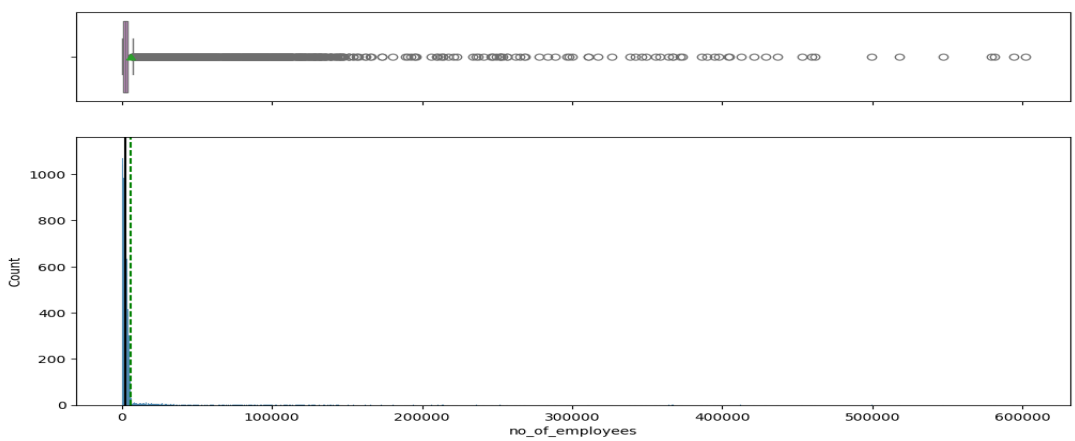

Figure 1(no. of employees)

- The "no of employees" attribute is right skewed distribution with a minimum of 11 and maximum of 60,000 plus
- The graph shows that no of employees above 20,000 are above the 75% quartile range

- The boxplot highlights a compact interquartile range and several outlier markers that pull the upper whisker upward.

Observations on prevailing wage

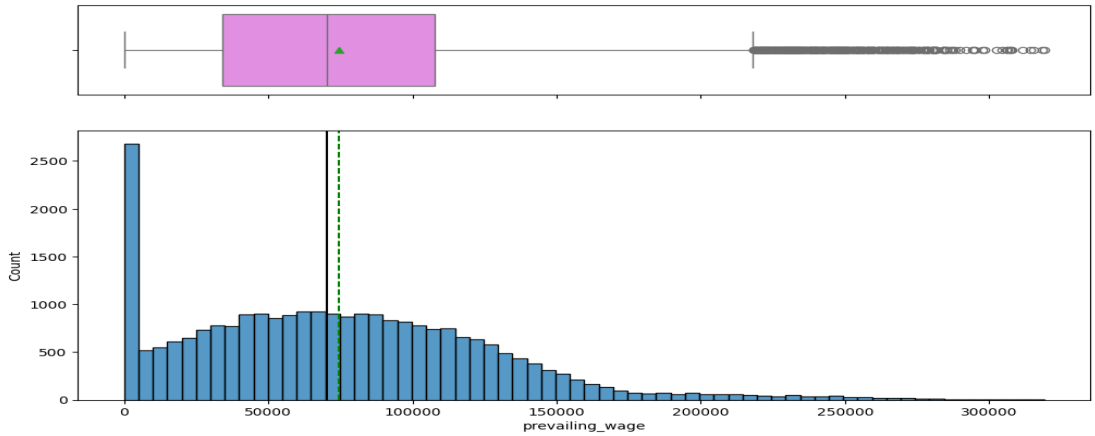

Figure 2(prevailing wage)

- The "no of employees" attribute is right skewed distribution with a minimum of 11 and maximum of 60,000 plus
- The graph shows that no of employees above 20,000 are above the 75% quartile range
- The boxplot highlights a compact interquartile range and several outlier markers that pull the upper whisker upward.

Observations on continent

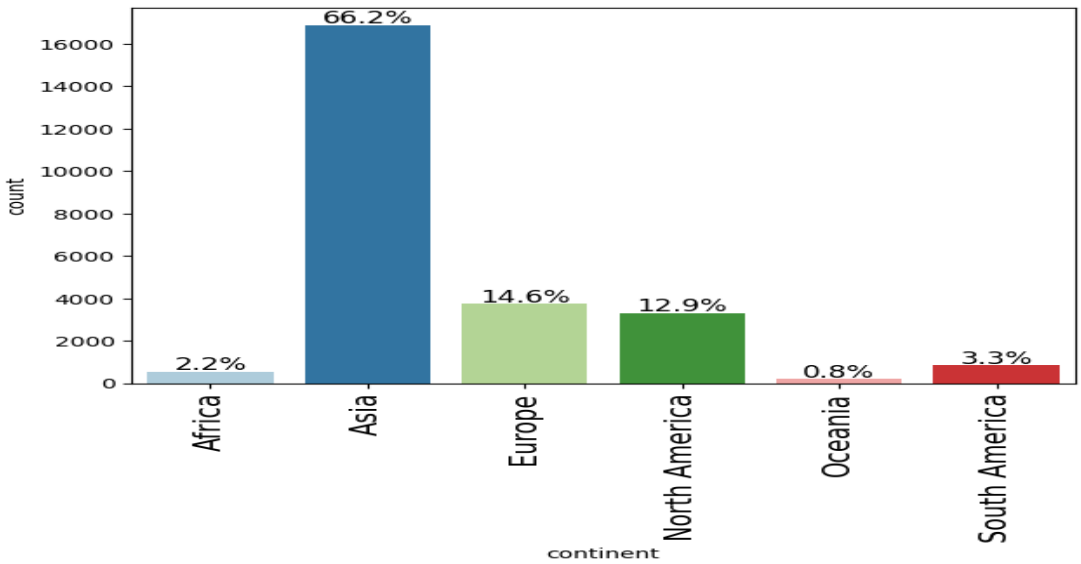

Figure 3(continents)

- Asia has the maximum number of applicants followed by Europe and North America.
- Nearly 66.2% of the application are from Asia.
- Oceania has the least at 0.8%.

Observations on education of employee

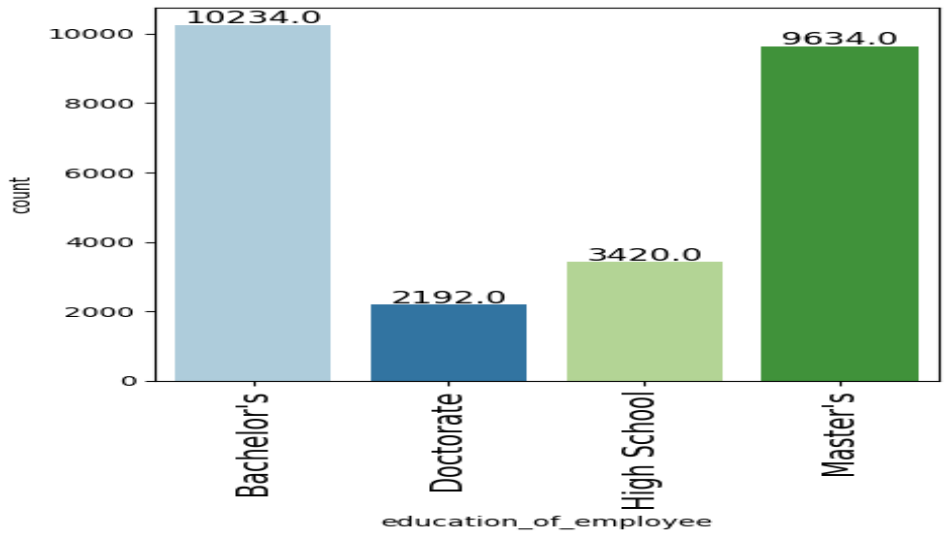

Figure 4(education of employees)

- The graph is characterized by two dominant, tall bars (Bachelor's and Master's) with the categories exhibiting a descending order of heights.
- Bachelor's is the highest and doctor is the lowest.

Observations on job experience

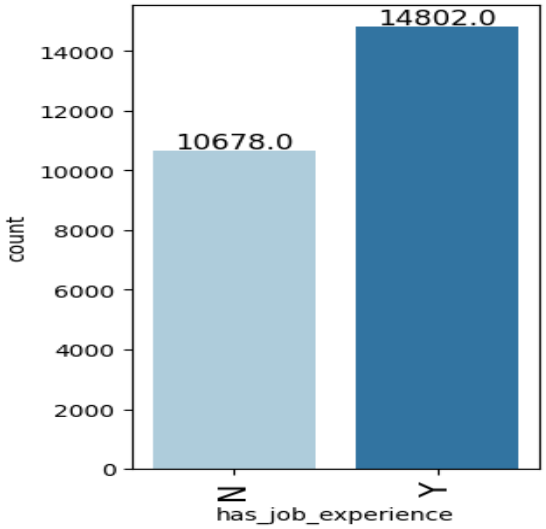

Figure 5(job experience)

- Maximum number of people have job experience.
- Nearly 58% of the employees who apply for VISA have job experience.
- The number of employees who don't have job experience is 42%, so there is a big difference but it isn't completely overshadowed.

Observations on job training

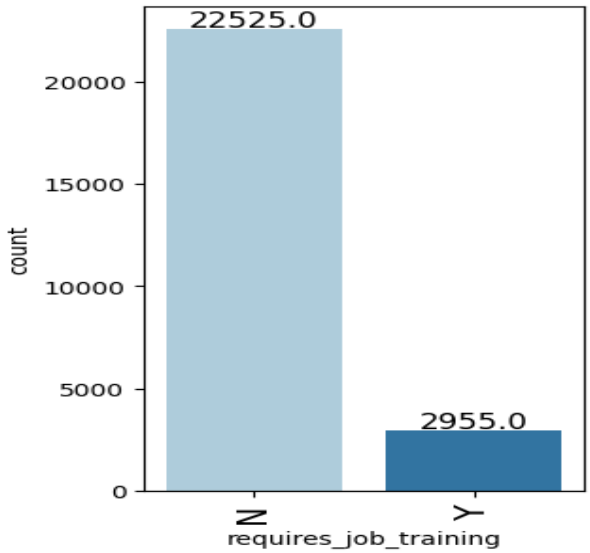

Figure 6(require job training)

- Nearly 88% of the employees don't need any job training. This can be an important factor for the employer to make decision.
- Only 11.6% need job training.

Observations on region of employment

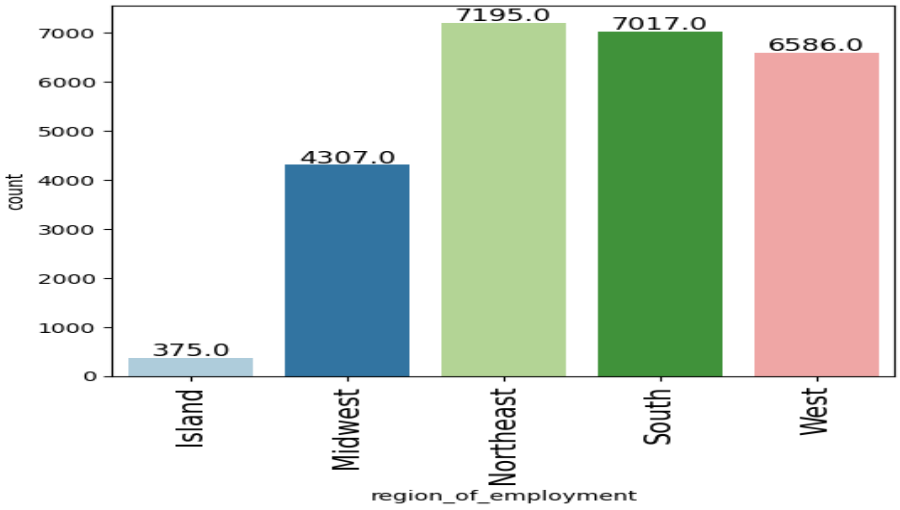

Figure 7(region of employment)

- Northeast, south and west have the maximum number of employments with northeast being the maximum at 28.2%.
- Midwest has decent number of employments at nearly 17%.
- Islands have the least at 1.5%.

Observations on unit of wage

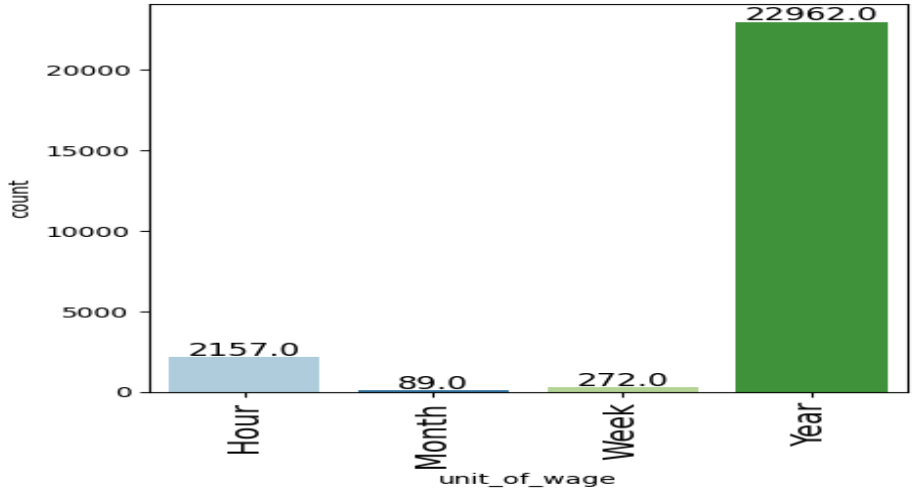

Figure 8(unit of wage)

- 90% of employees get an annual package .
- 8.5% of them work on hourly basis.
- Month and week units of wage are the least .

Observations on case status

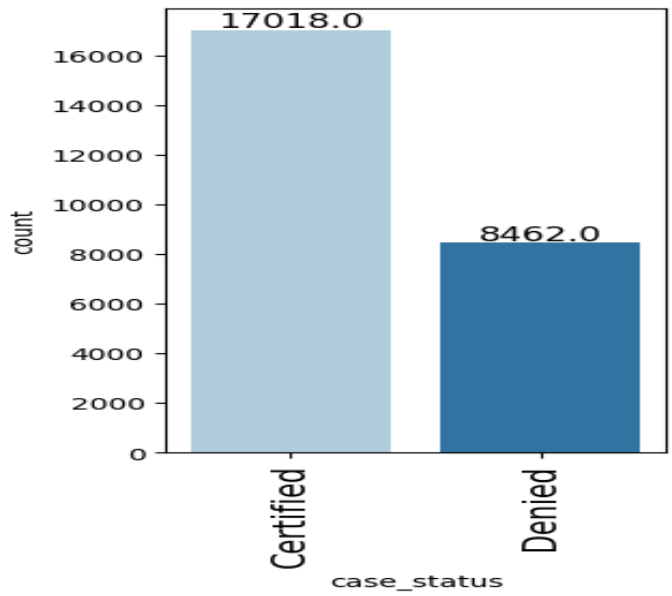

Figure 9(case status)

- 66.8% of the cases are certified.
- Around 33.2% of the cases are denied.

Bivariate Analysis

Confusion Matrix

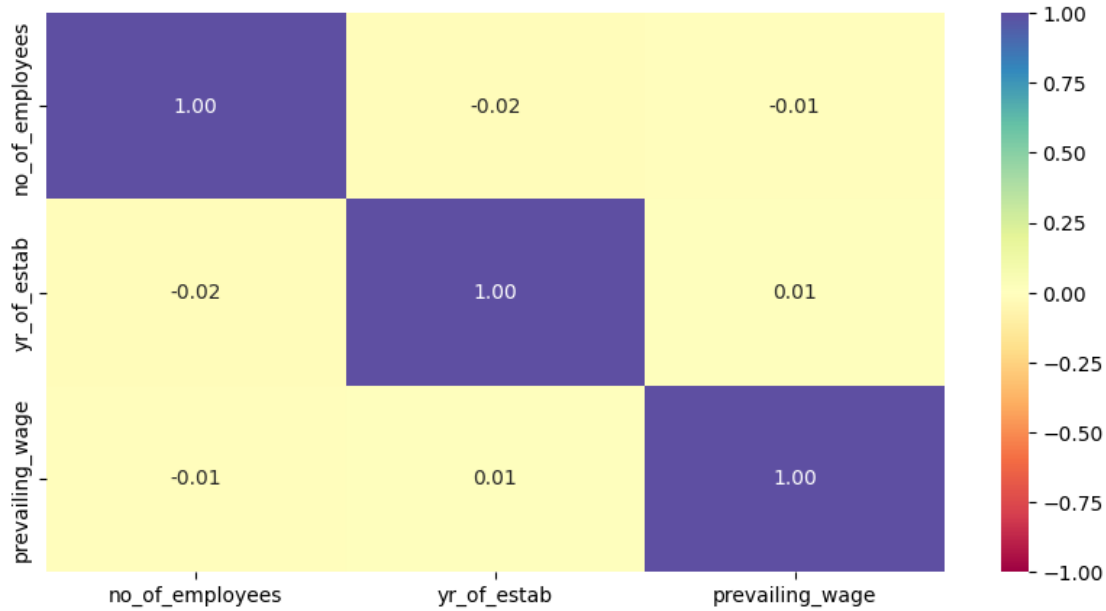

Figure 10(confusion matrix)

- There seems to be no correlation between any of the numerical features.

educations impact on visa certification

case_status	Certified	Denied	All
education_of_employee			
All	17018	8462	25480
Bachelor's	6367	3867	10234
High School	1164	2256	3420
Master's	7575	2059	9634
Doctorate	1912	280	2192

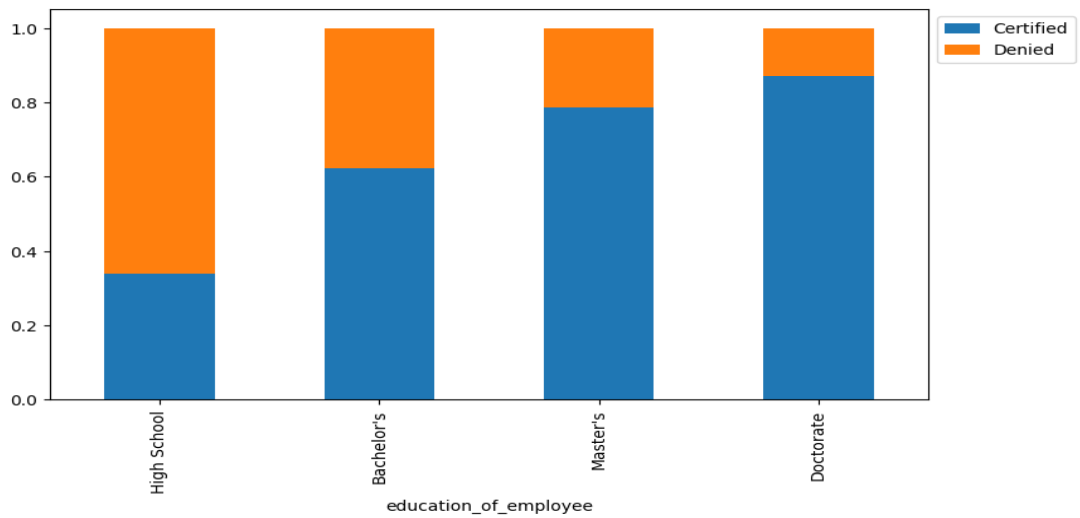

Figure 11(edu vs visa)

- Education seems to have a clear and direct correlation to the probability of the case being certified

Different regions vs educational backgrounds

Figure 12(region vs edu)

- Most regions seem to have the strongest preference towards Bachelor's except island and the Midwest where the prefer Master's.
- Masters is close behind followed distantly by high school and then doctorate.

Percentage of visa certifications across each region

case_status	Certified	Denied	All
region_of_employment			
All	17018	8462	25480
Northeast	4526	2669	7195
West	4100	2486	6586
South	4913	2104	7017
Midwest	3253	1054	4307
Island	226	149	375

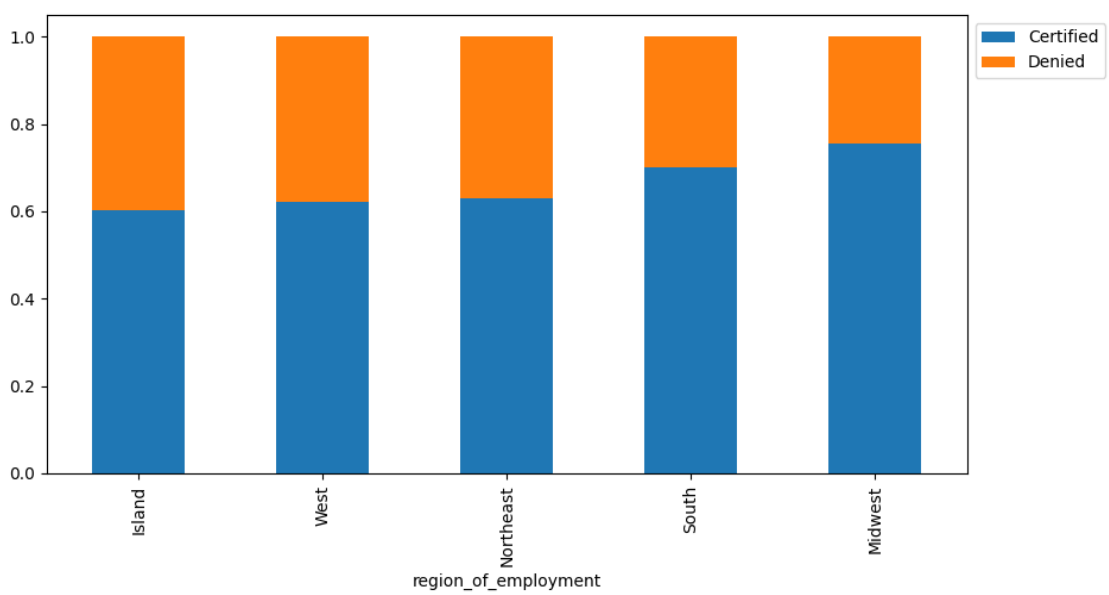

Figure 13(region vs visa)

- There seems to be a similar ratio between Certified/Denied among all regions, at around, 7:3, except South and Midwest where it is more in favor of Certified at around 8:2.

visa status variation across different continents.

case_status	Certified	Denied	All
continent			
All	17018	8462	25480
Asia	11012	5849	16861
North America	2037	1255	3292
Europe	2957	775	3732
South America	493	359	852
Africa	397	154	551
Oceania	122	70	192

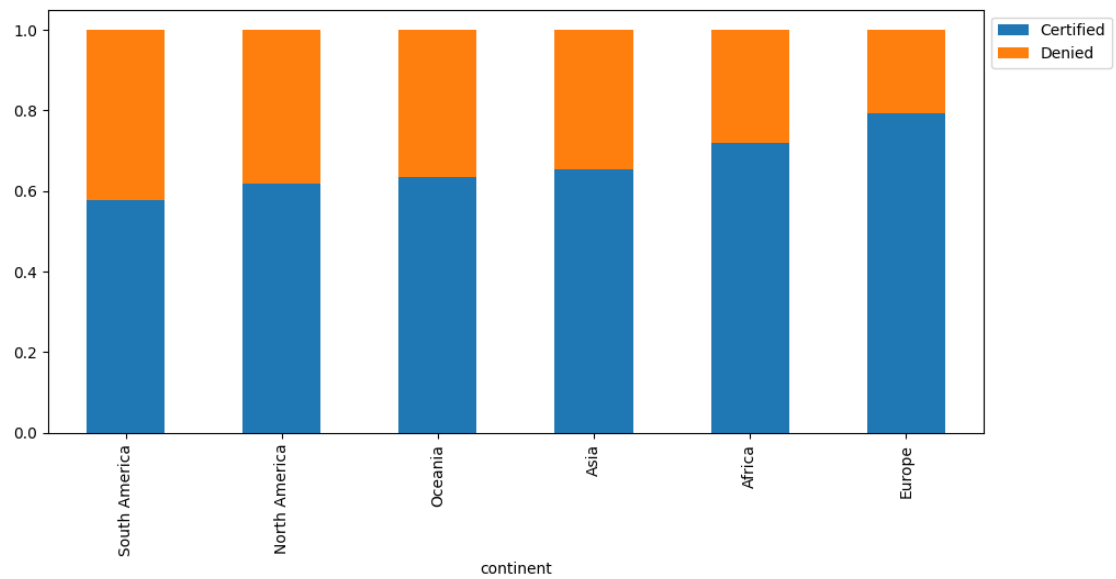

Figure 14(continent vs visa)

- Europe has the highest number of visas certified followed by Africa and Asia.
- Asia has the highest number of cases, but the percentage of visas getting certified is more for Europe.

work experience vs visa certification

case_status	Certified	Denied	All
has_job_experience			
All	17018	8462	25480
N	5994	4684	10678
Y	11024	3778	14802

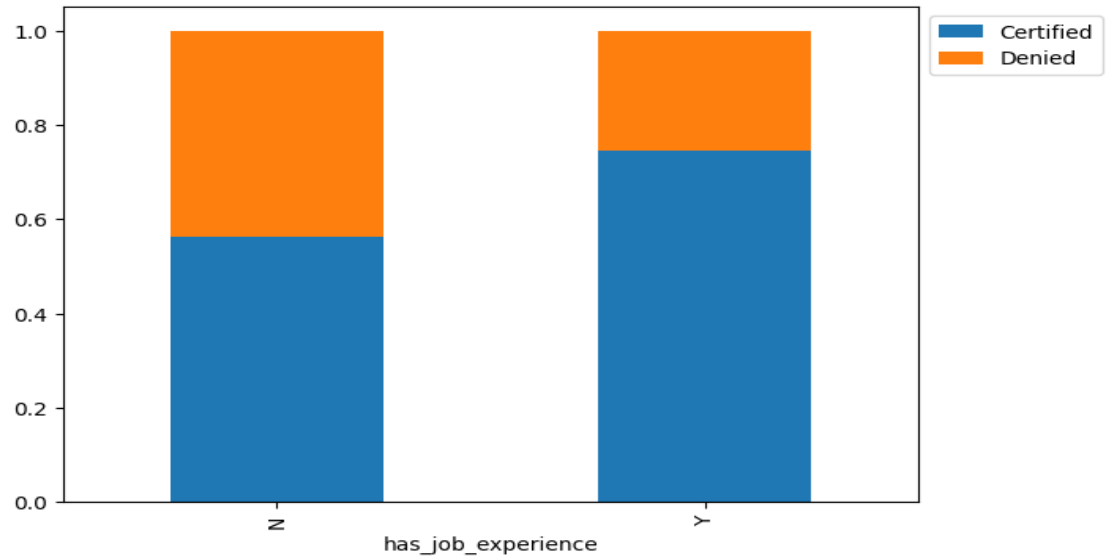

Figure 15(work ex vs visa)

- Employees with Experience have higher chances of getting Visa certified compared to the employees with no experience.

Do the employees who have prior work experience require any job training?

requires_job_training	N	Y	All
has_job_experience			
All	22525	2955	25480
N	8988	1690	10678
Y	13537	1265	14802

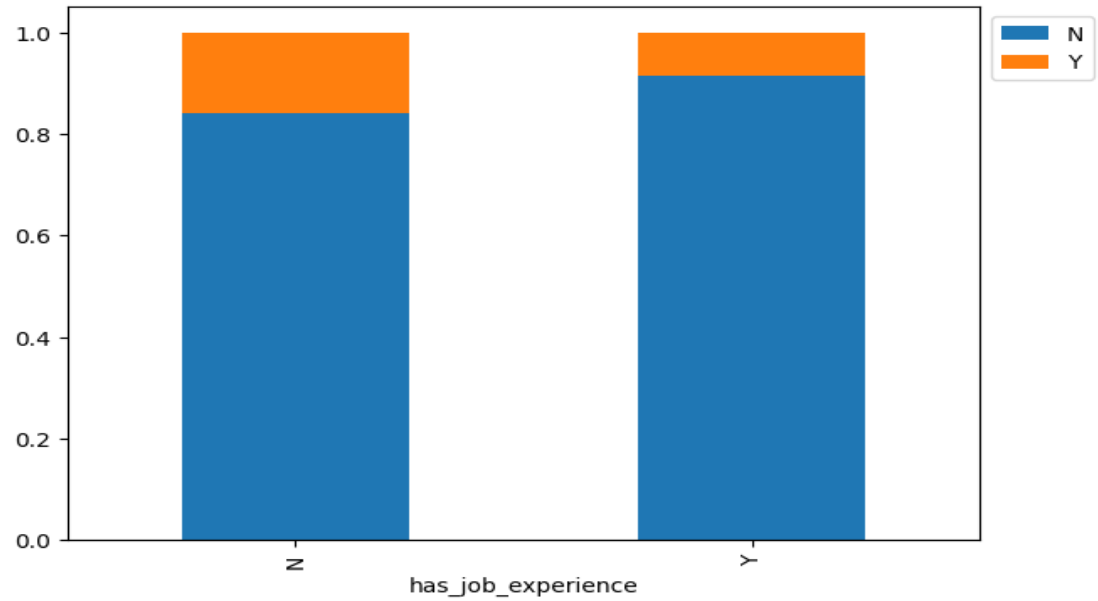

Figure 16(work ex vs training)

- Looks like Employees with or without experience have low requirement for job training.

prevailing wage vs visa status

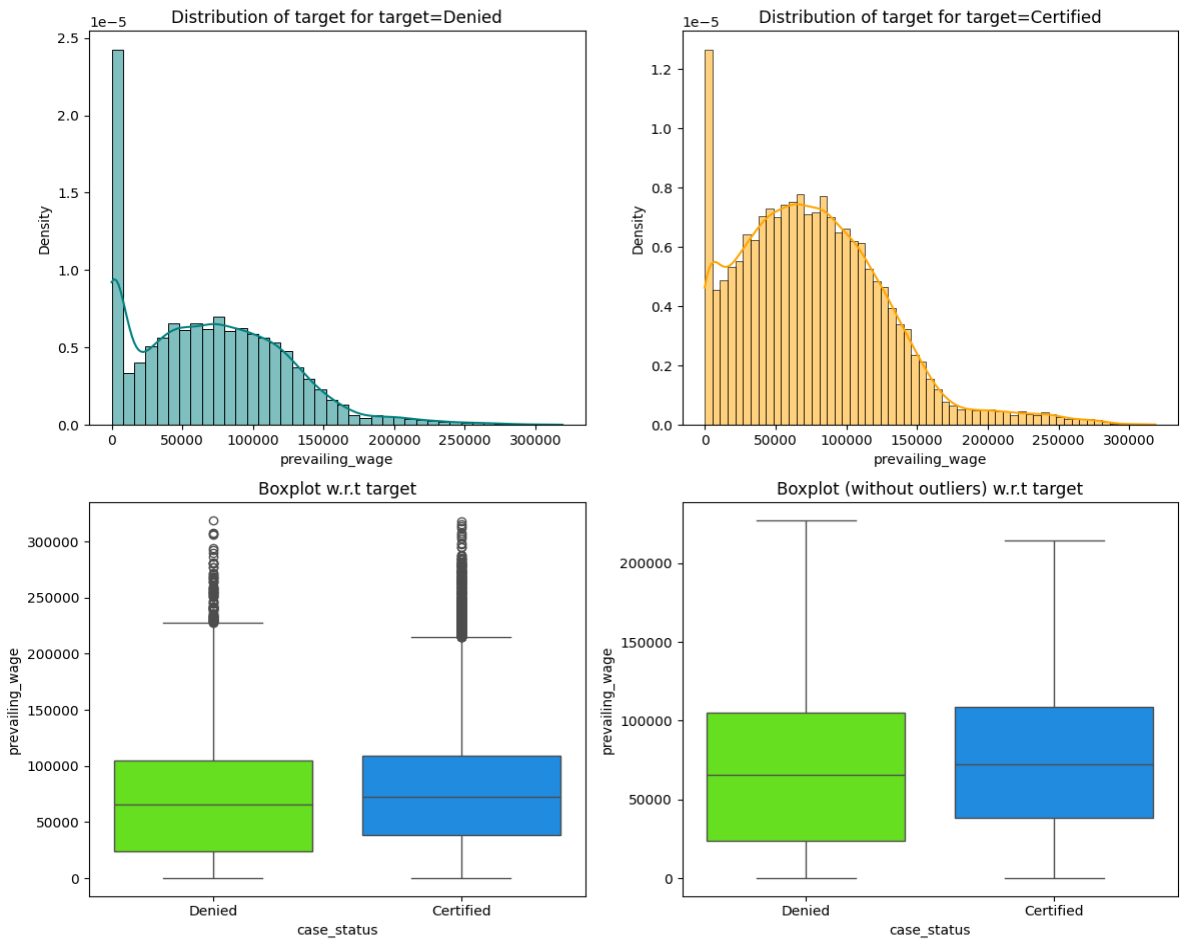

Figure 17(wage vs visa)

- As the prevailing wage increases , there is a slightly higher chance of getting the visa certified.
- The plot after removing the **outliers** clearly shows that there is a better chance of getting the visa certified with higher prevailing wage

Checking if the prevailing wage is similar across all the regions of the US

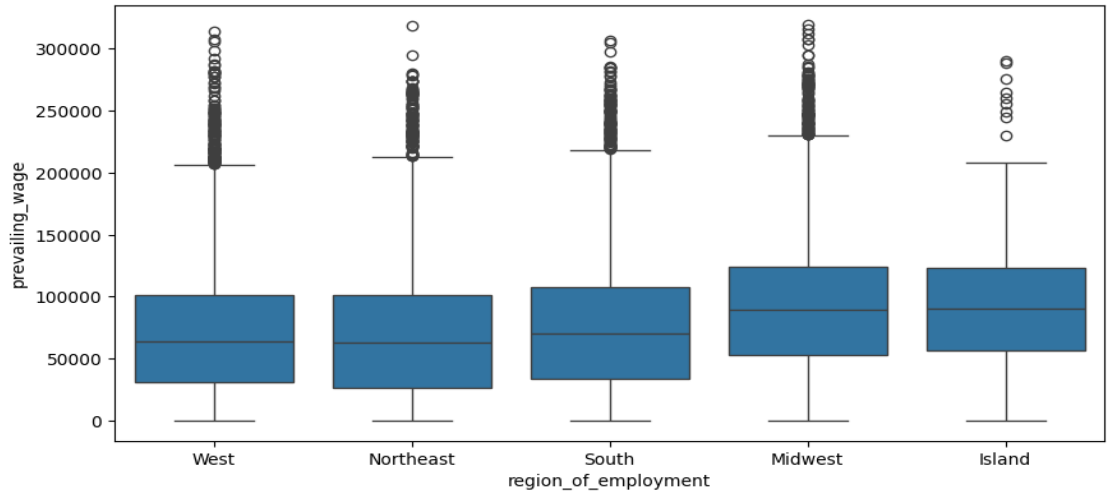

Figure 18(wage vs region)

- Prevailing wage is higher in the Midwest and Island regions compared to other regions of the US.

different units (Hourly, Weekly, etc) vs visa status

```
stacked_barplot(data, "unit_of_wage", "case_status")
```

```
case_status   Certified  Denied    All
unit_of_wage
All               17018    8462  25480
Year              16047    6915  22962
Hour                747    1410   2157
Week                169     103    272
Month                55      34     89
```

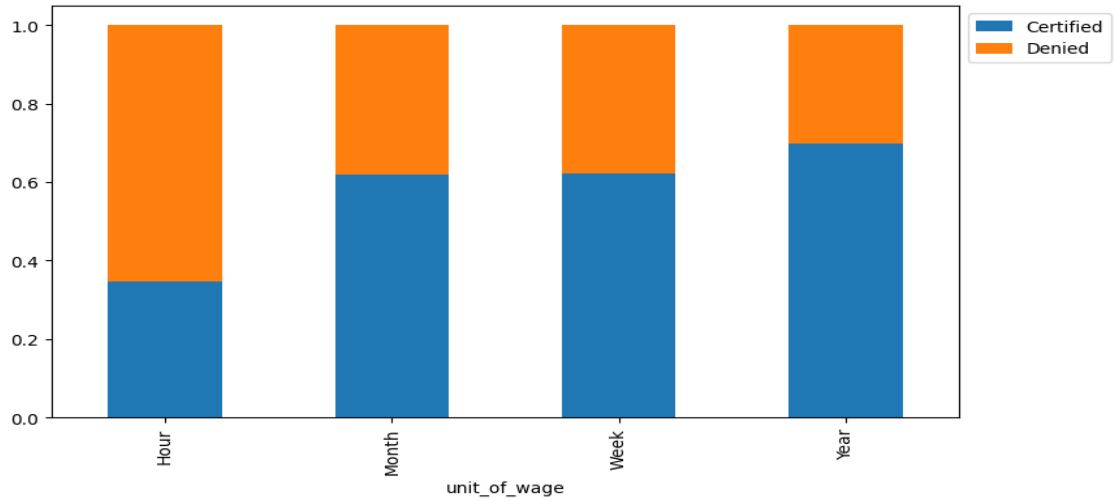

Figure 19(unit of wage vs visa)

- There is higher chance of getting visa certified when the prevailing wage has the unit of wage as year, month or week as compared to hour.

Outlier Check

- Let's check for outliers in the data.

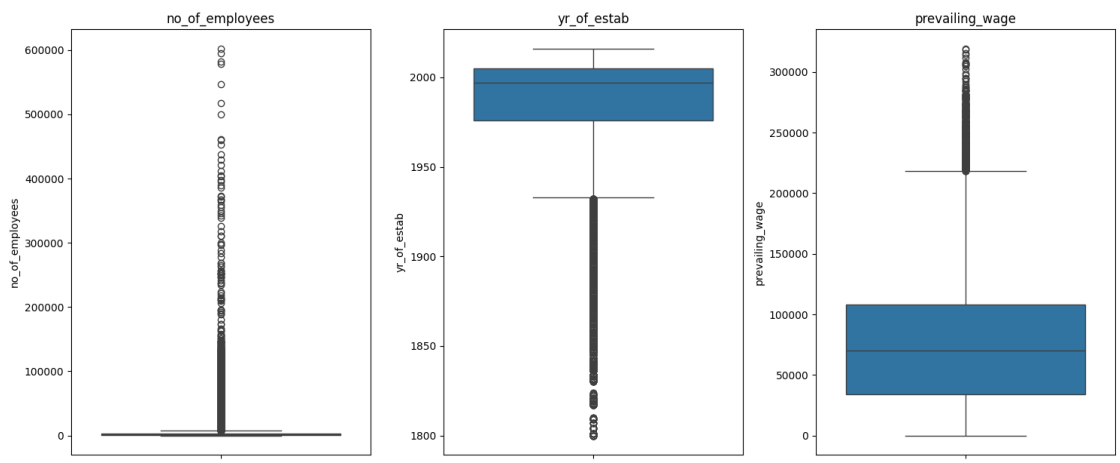

Figure 20(outliers check)

Data Preparation for modeling

- We want to predict which visa will be certified.
- Before we proceed to build a model, we'll have to encode categorical features.
- We'll split the data into train and test to be able to evaluate the model that we build on the train data.

```
Shape of Training set :  (17836, 21)
Shape of test set :  (7644, 21)
Percentage of classes in training set:
case_status
1    0.667919
0    0.332081
Name: proportion, dtype: float64
Percentage of classes in test set:
case_status
1    0.667844
0    0.332156
Name: proportion, dtype: float64
```

Model evaluation criterion

Model can make wrong predictions as:

1. Model predicts that the visa application will get certified but in reality, the visa application should get denied.
2. Model predicts that the visa application will not get certified but in reality, the visa application should get certified.

Which case is more important?

- Both the cases are important as:
- If a visa is certified when it had to be denied a wrong employee will get the job position while US citizens will miss the opportunity to work on that position.
- If a visa is denied when it had to be certified the U.S. will lose a suitable human resource that can contribute to the economy.

How to reduce the losses?

- `F1 Score` can be used a the metric for evaluation of the model, greater the F1 score higher are the chances of minimizing False Negatives and False Positives.
- We will use balanced class weights so that model focuses equally on both classes.
- The model_performance_classification_sklearn function will be used to check the model performance of models.
- The confusion_matrix_sklearn function will be used to plot the confusion matrix.

Decision Tree Model

Checking model performance on training set

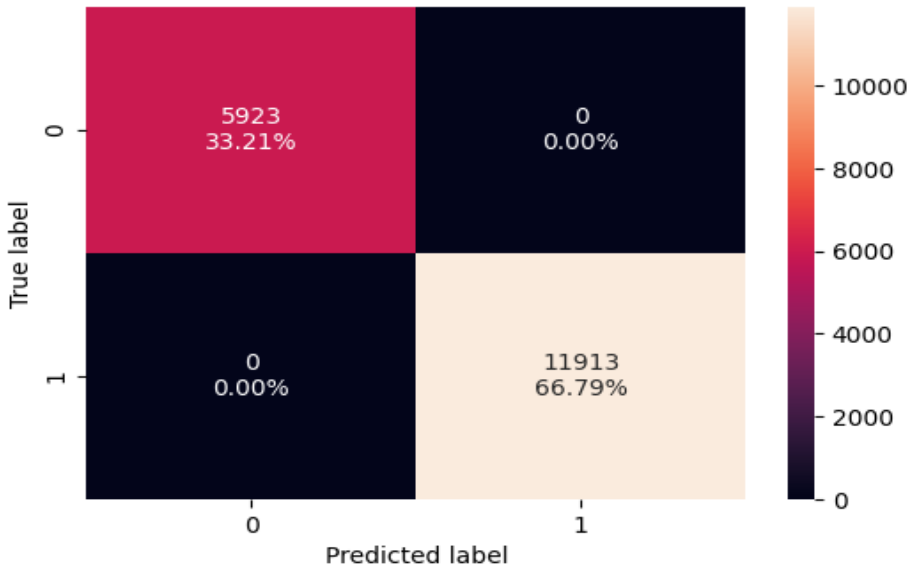

Figure 21(dec. tree performance train)

accuracy	Recall	Precision	F1
1.0	1.0	1.0	1.0

Checking model performance on test set

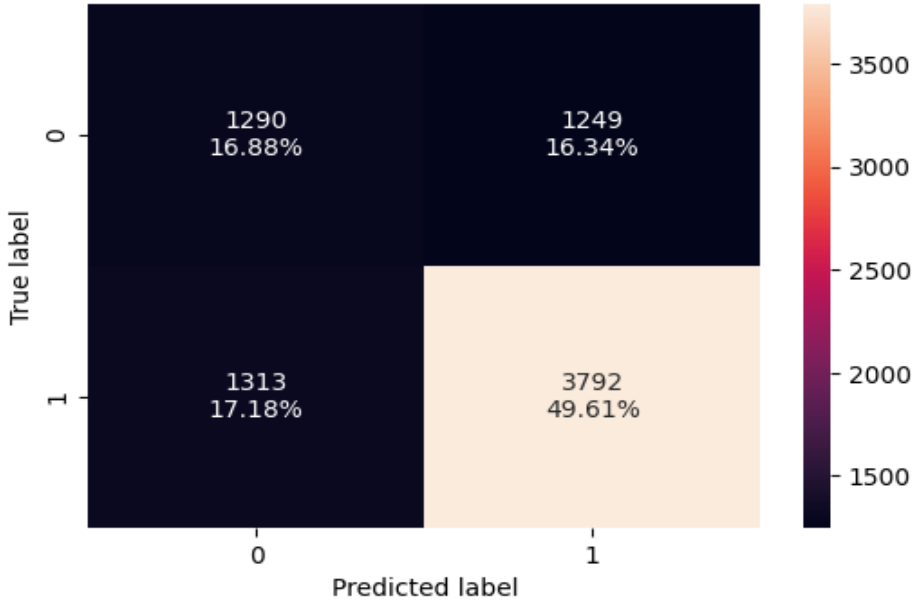

Figure 22(dec. tree performance test)

accuracy	Recall	Precision	F1
0.664835	0.742801	0.75223	0.74748

- The performance of the test data is very poor with a F1 score of 0.74748.
- Let's do hyperparameter tuning and check for the model performance.

Hyperparameter Tuning - Decision Tree performance

Training set

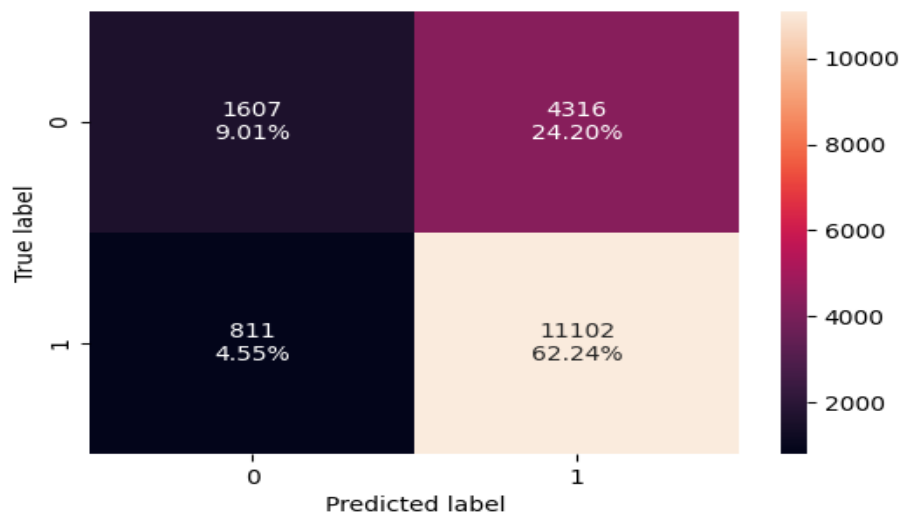

Figure 23(dec. tree performance tuned train)

accuracy	Recall	Precision	F1
0.7125	0.9319	0.7200	0.8124

Testing set

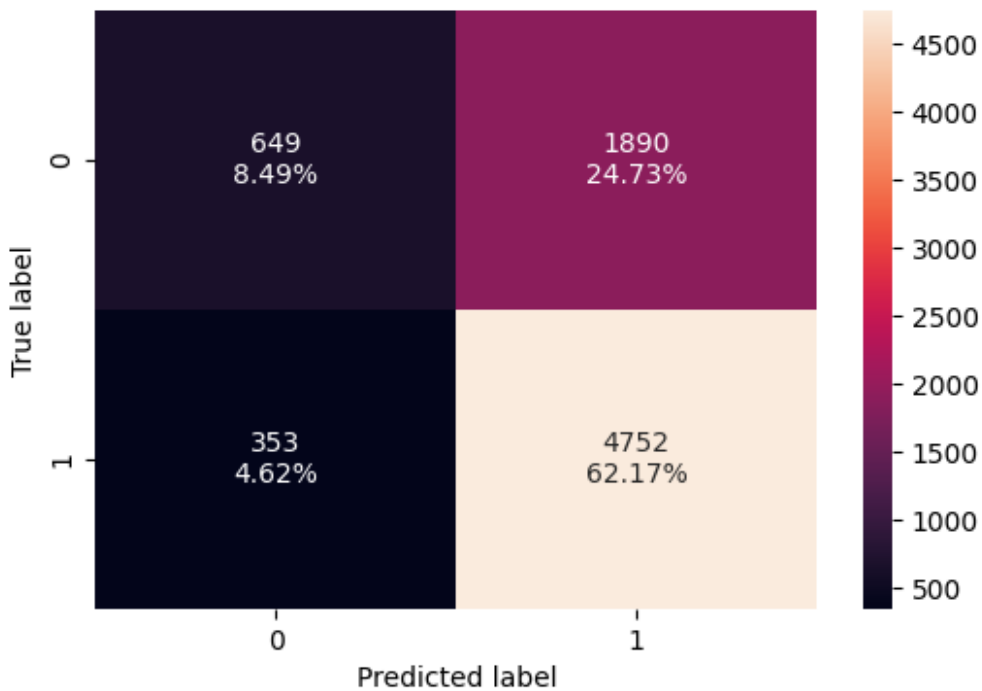

Figure 24(dec. tree performance tuned test)

accuracy	Recall	Precision	F1
0.7065	0.9308	0.7154	0.8090

- The F1 score for the train data is decreased, but the performance on the test data is improved to F1 =0.8090.
- Overfitting is reduced.
- Decision tree classifier performs best at the max depth of 10, max leaf nodes of 2, min samples leaf of 3 with min impurity decrease of 0.0001 and class weight "balanced".

Bagging Classifier

Checking model performance on training set

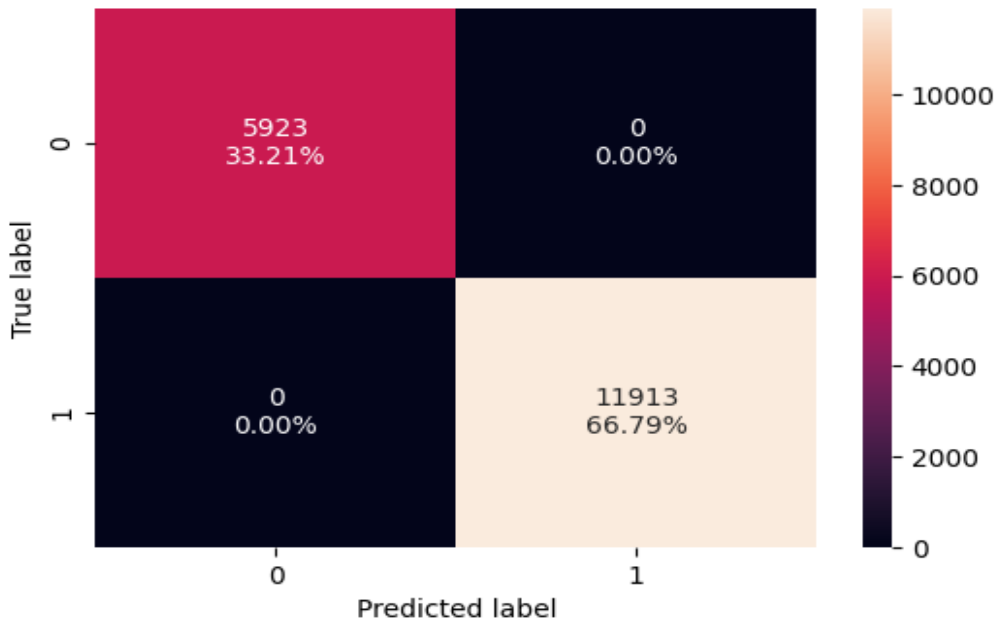

Figure 25(bagging cls train)

accuracy	Recall	Precision	F1
0.9851	0.9859	0.99181	0.9888

Checking model performance on test set

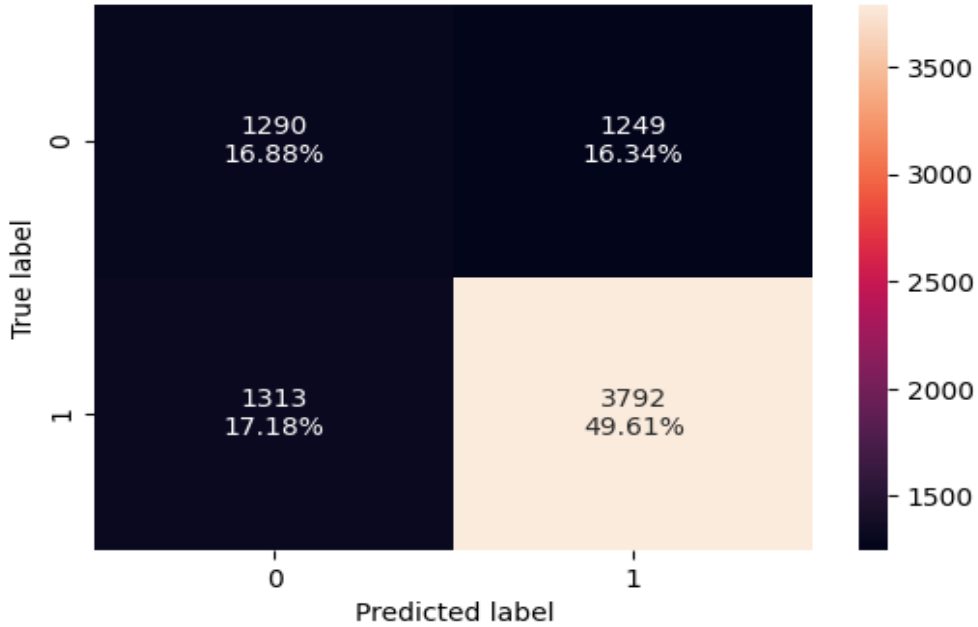

Figure 26(bagging cls test)

accuracy	Recall	Precision	F1
0.6915	0.7641	0.7717	0.76791

- Performance difference on training and testing data is too due to overfitting
- Let's check if the performance will improve with hyperparameter tuning.

Hyperparameter Tuning - Bagging Classifier

Checking model performance on training set

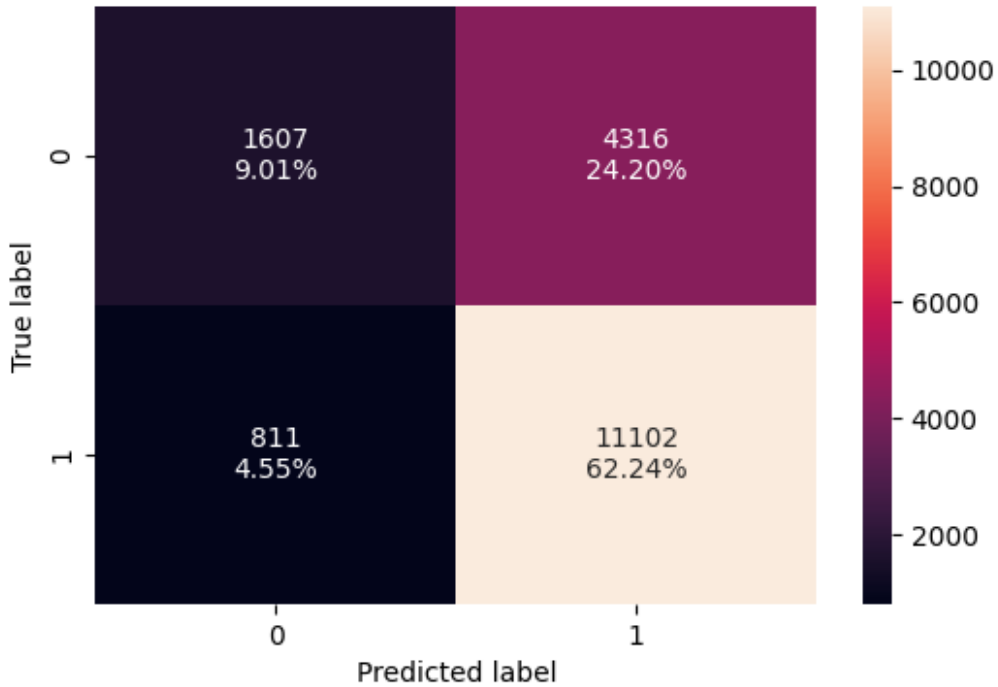

Figure 27(bagging cls tuned train)

accuracy	Recall	Precision	F1
0.9961	0.9999	0.9944	0.997154

Checking model performance on test set

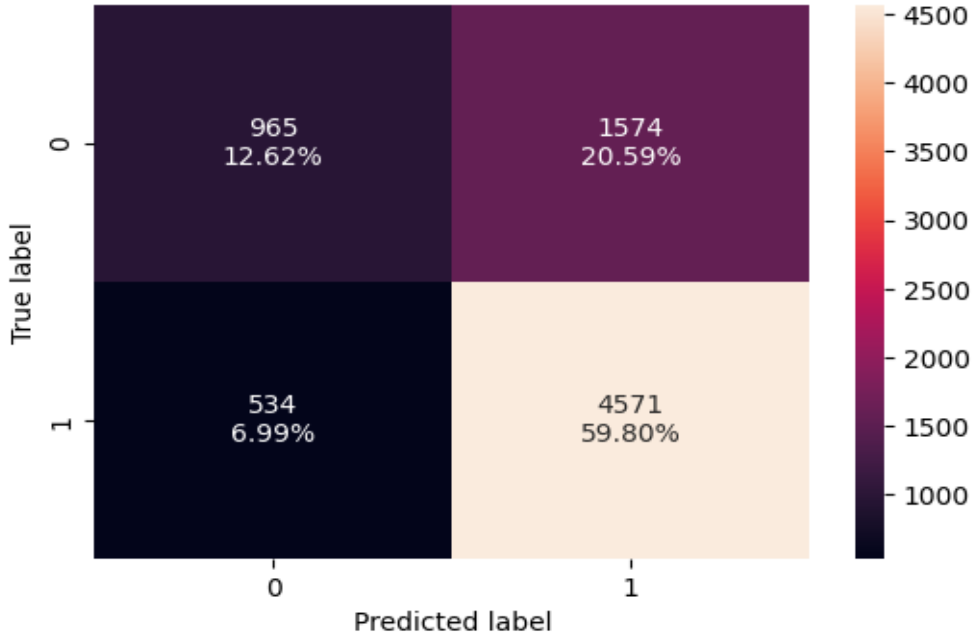

Figure 28(bagging cls tuned test)

accuracy	Recall	Precision	F1
0.7242	0.8953	0.74385	0.81262

There is a performance improvement but it still seem that training data is overfit.

Random Forest

Checking model performance on training set

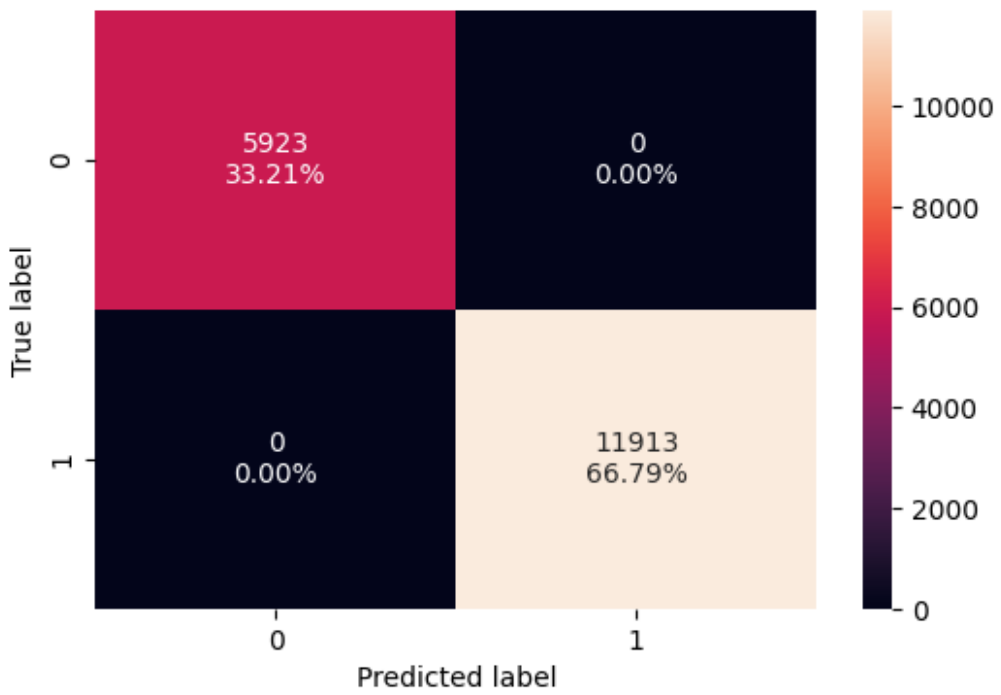

Figure 29(Random forest train)

accuracy	Recall	Precision	F1
0.99994	0.99916	1.0	0.99958

Checking model performance on test set

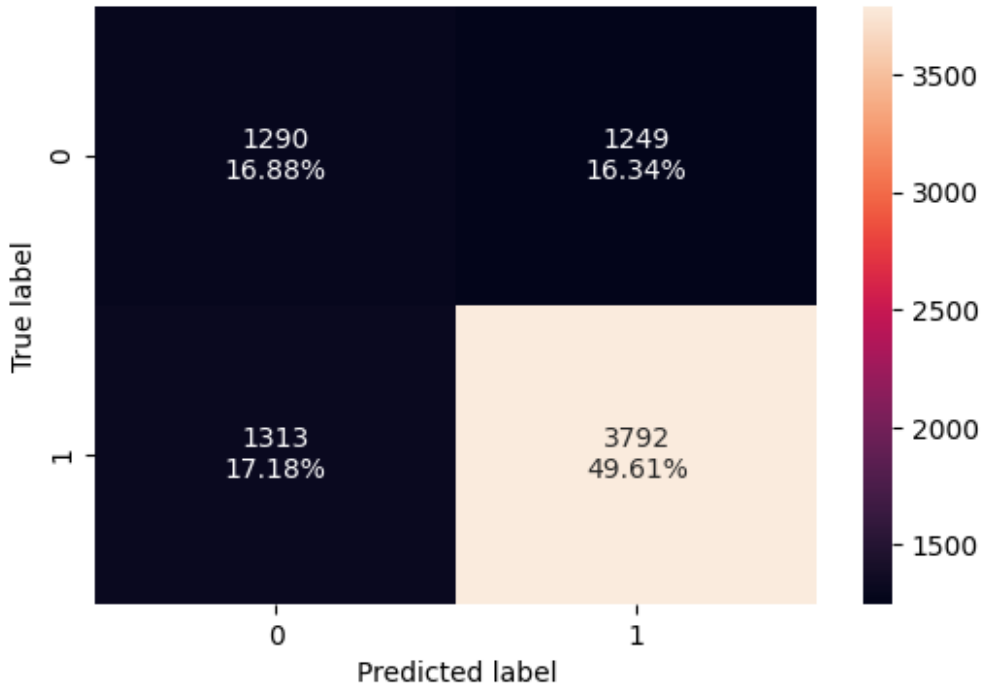

Figure 30(Random forest test)

accuracy	Recall	Precision	F1
0.720827	0.832125	0.768886	0.79924

- Random Forest performs well on the Train data with F1=0.99958 but comparatively poor on test data with F1=0.79924. The model is overfitting.
- The confusion Matrix of the test data shows false positives and false negatives.

Hyperparameter Tuning - Random Forest

Checking model performance on training set

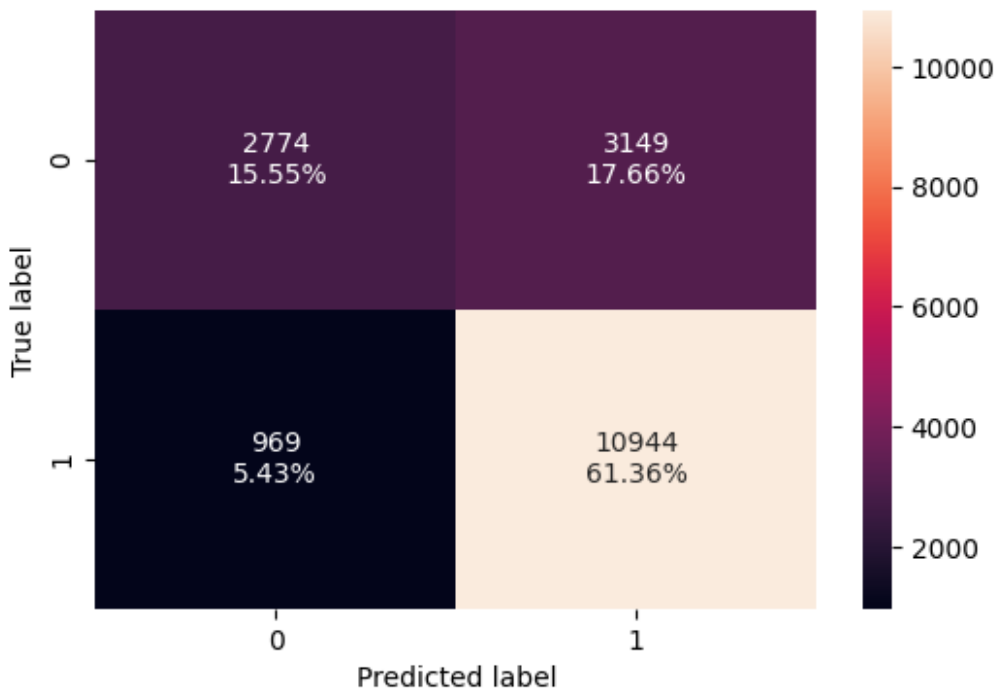

Figure 31(Random forest tuned train)

accuracy	Recall	Precision	F1
0.76911	0.91866	0.77655	0.84165

Checking model performance on test set

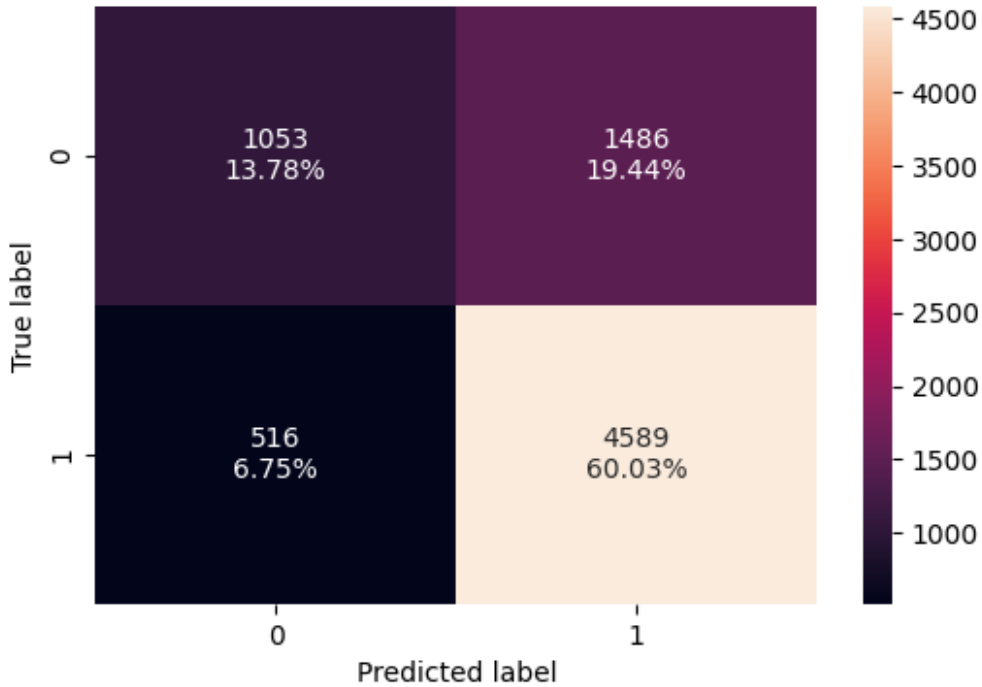

Figure 32(Random forest tuned test)

accuracy	Recall	Precision	F1
0.73809	0.89892	0.75539	0.82093

- The model performance on the train data is not overfitting .
- The F1 score of both train and test are almost close to each other.
- Random forest classifier shows the best fit model with max depth =10, max features ='sqrt', min sample split=7, estimators=20, oob score=True and random state 1 .

AdaBoost Classifier

Checking model performance on training set

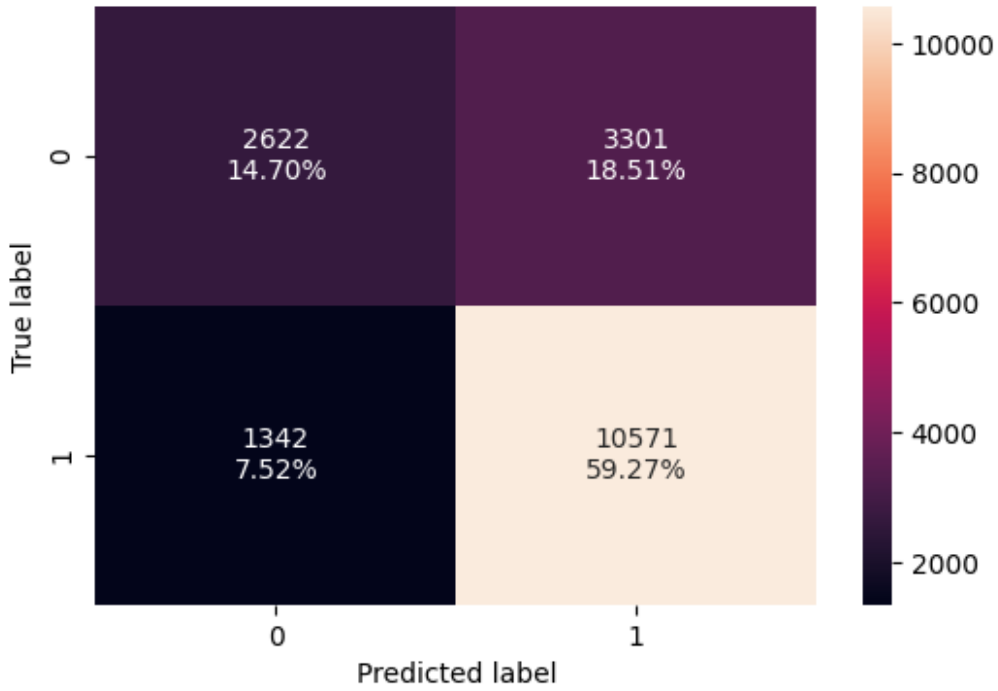

Figure 33(adaboost train)

accuracy	Recall	Precision	F1
0.73822	0.88718	0.76068	0.81908

Checking model performance on test set

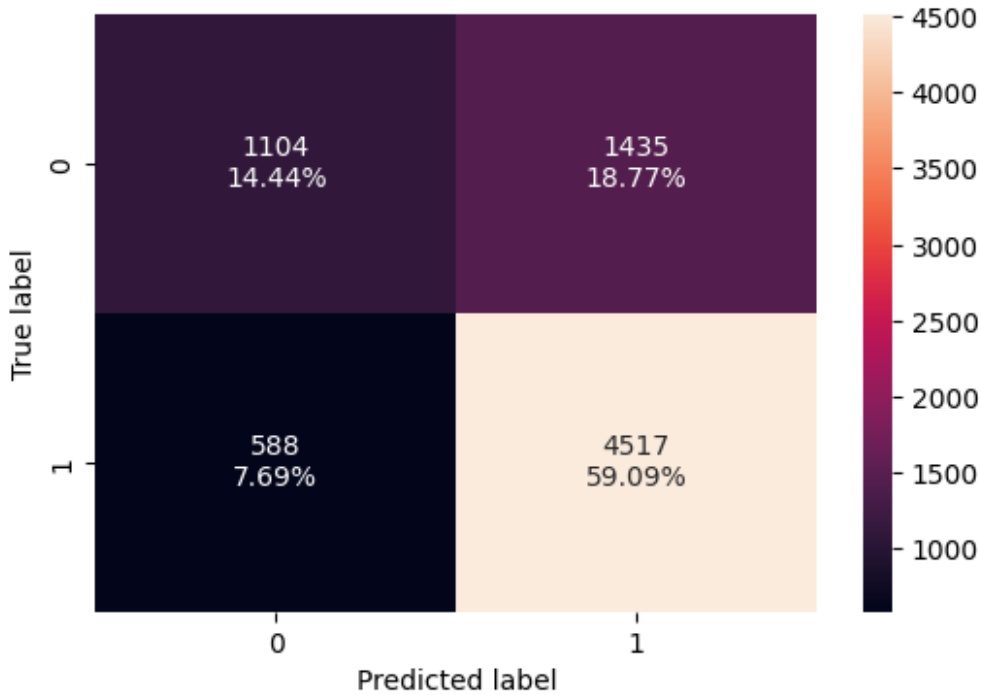

Figure 34((adaboost test)

accuracy	Recall	Precision	F1
0.73430	0.885015	0.75779	0.81648

- The performance of the model is better compared to other models, the F1 =0.81648.The model is also not overfitting.
- Performance on both the train and test data is almost close. Let's check if we can improve the F1 score by hyperparameter tuning.

Hyperparameter Tuning - AdaBoost Classifier

Checking model performance on training set

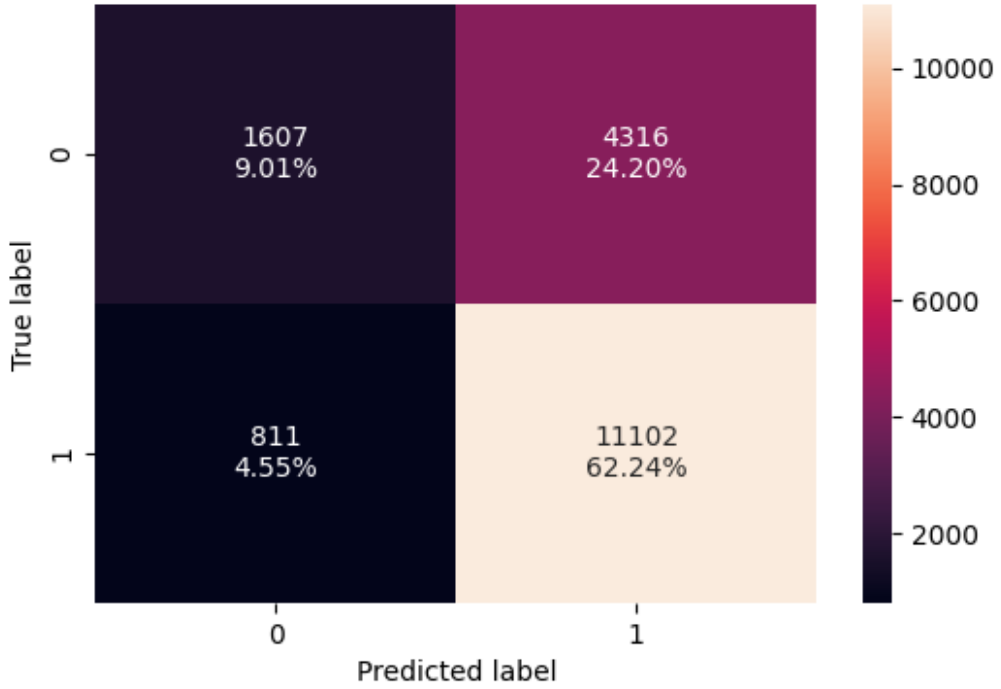

Figure 35(adaboost tuned train)

accuracy	Recall	Precision	F1
0.719163	0.78141	0.79469	0.78799

Checking model performance on test set

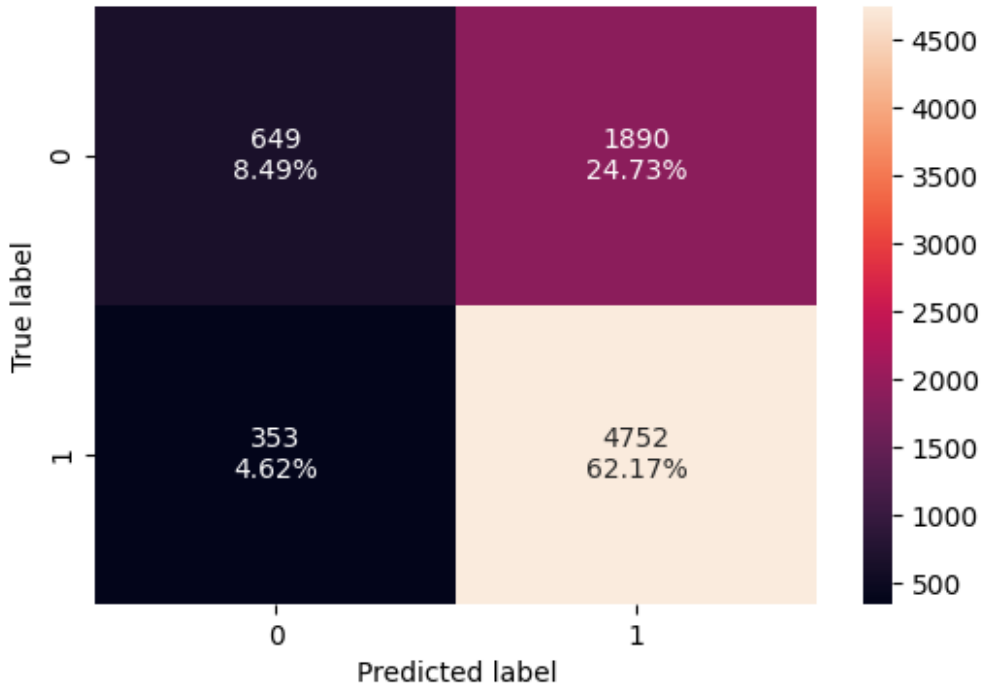

Figure 36(adaboost tuned test)

accuracy	Recall	Precision	F1
0.71664	0.78158	0.79151	0.78651

- Performace has reduced after hyperparameter tuning.

Gradient Boosting Classifier

Checking model performance on training set

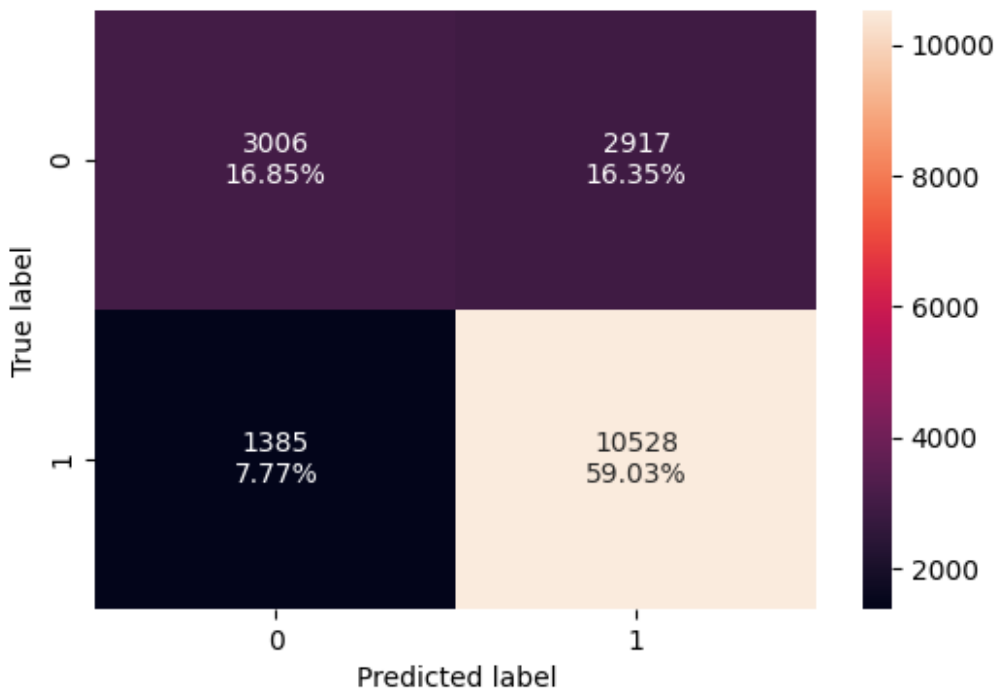

Figure 37(gradient boost train)

accuracy	Recall	Precision	F1
0.75880	0.88374	0.783041	0.83034

Checking model performance on test set

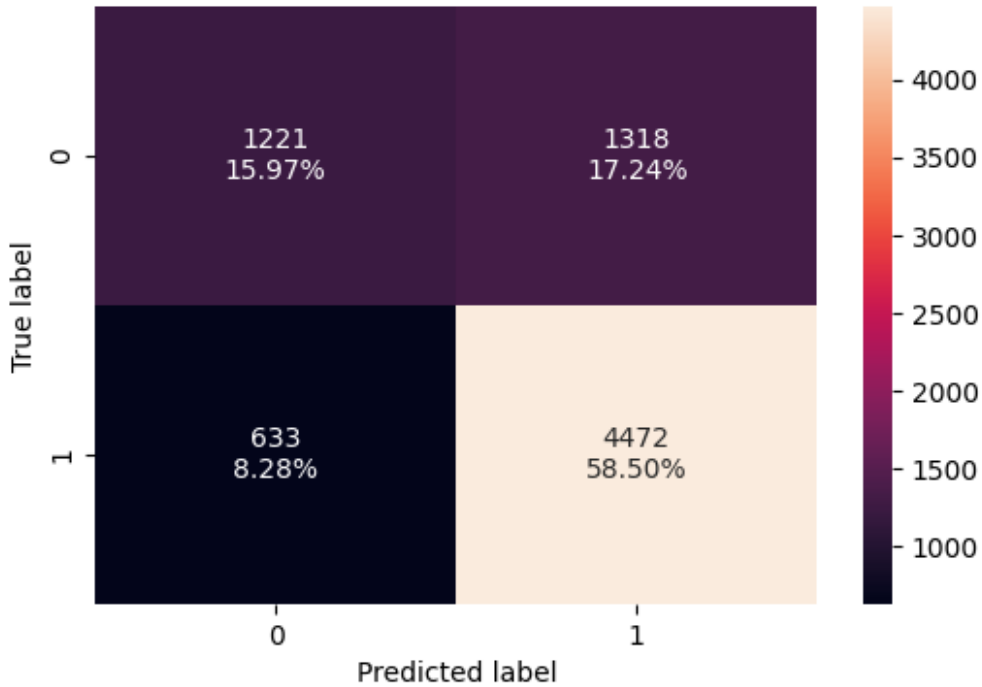

Figure 38(gradient boost test)

accuracy	Recall	Precision	F1
0.744767	0.87600	0.77236	0.82092

- Classifier gives a F1 =0.8303 on the train data and F1=0.82092 on the test data, The performance is almost close and not overfitting.
- Let's check if hyperparameter tuning can enhance the model performance.

Hyperparameter Tuning - Gradient Boosting Classifier

Checking model performance on training set

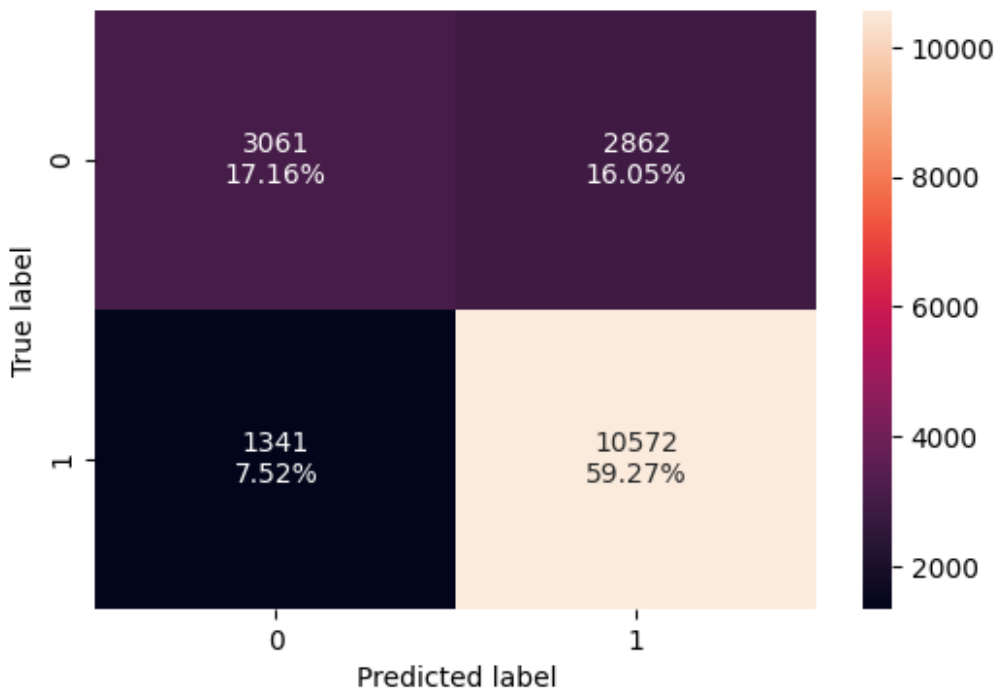

Figure 39(gradient boost tuned train)

accuracy	Recall	Precision	F1
0.76401	0.88264	0.78905	0.83323

Checking model performance on test set

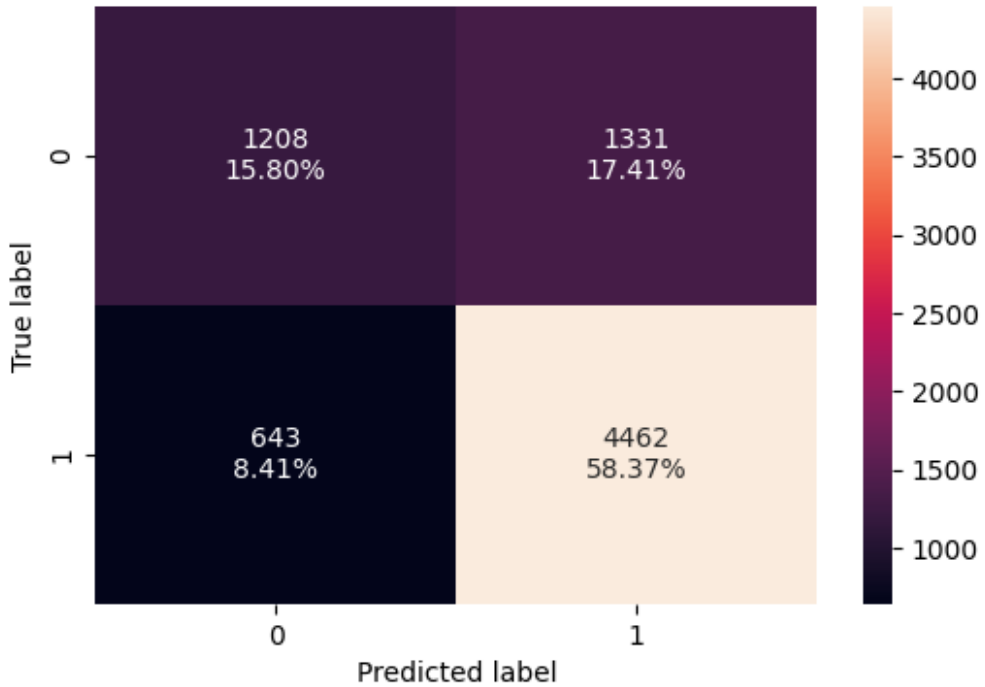

Figure 40(gradient boost tuned test)

accuracy	Recall	Precision	F1
0.74345	0.87130	0.77329	0.81937

- The F1 score on the test data doesn't seem to improve much with hyperparameter tuning.

XGBoost Classifier

Checking model performance on training set

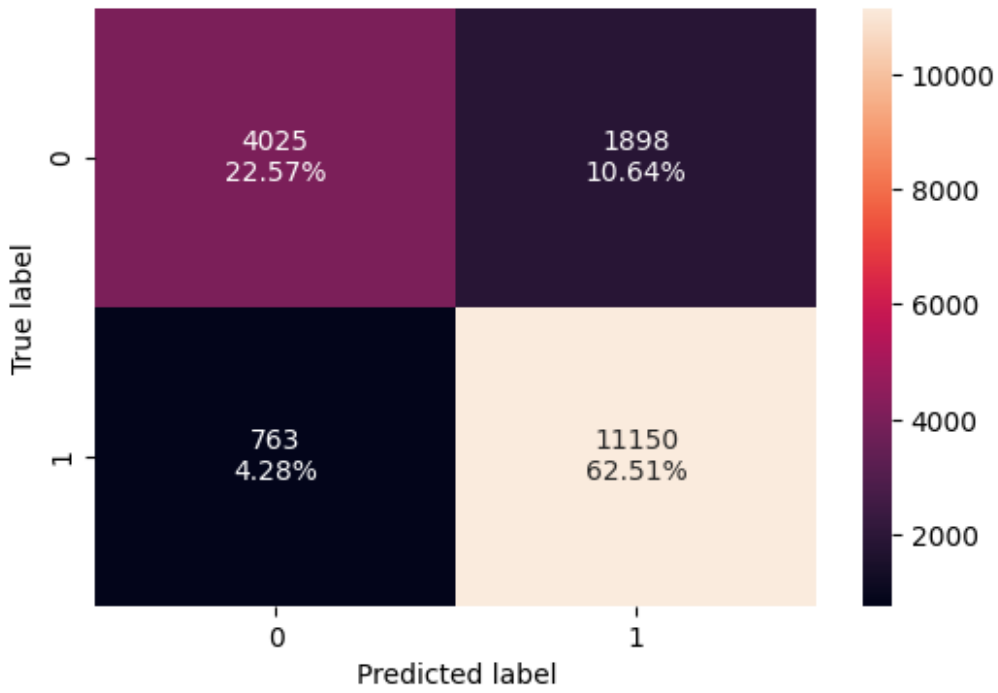

Figure 41(xgboost train)

accuracy	Recall	Precision	F1
0.83875	0.93141	0.843482	0.88527

Checking model performance on test set

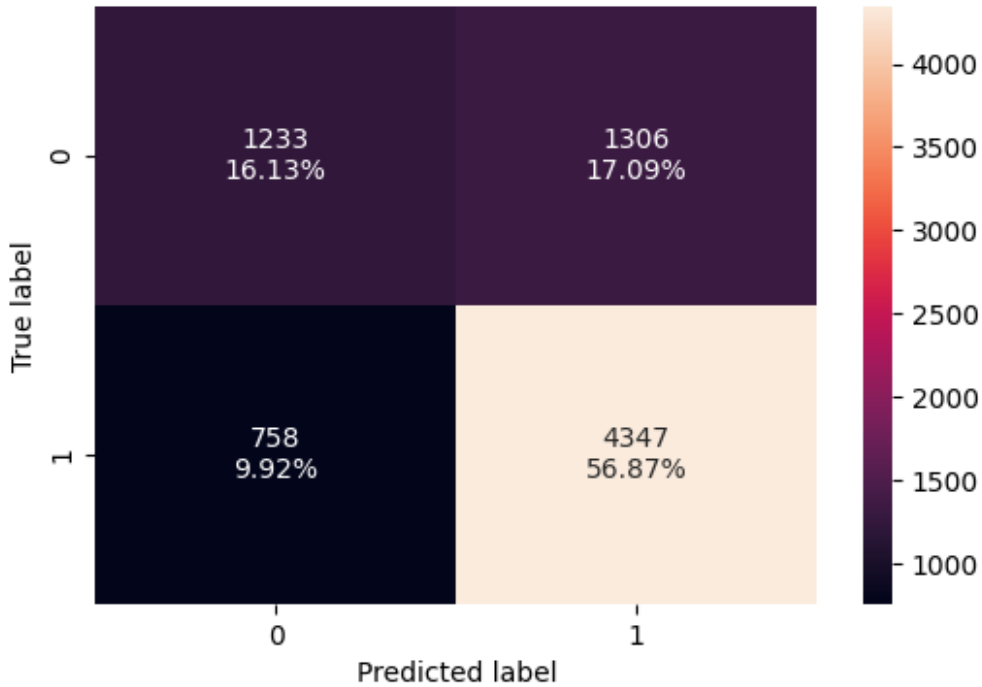

Figure 42(gradient boost test)

accuracy	Recall	Precision	F1
0.73325	0.86072	0.76791	0.81167

- We have defined the XG boost classifier with random state =1 and evaluation metric as **log loss.**
- The performance on the test data, F1= 0.8116 is comparatively less compared to that of the train data.
- Let's check the performance after hyperparameter tuning**.**

Hyperparameter Tuning - XGBoost Classifier

Checking model performance on training set

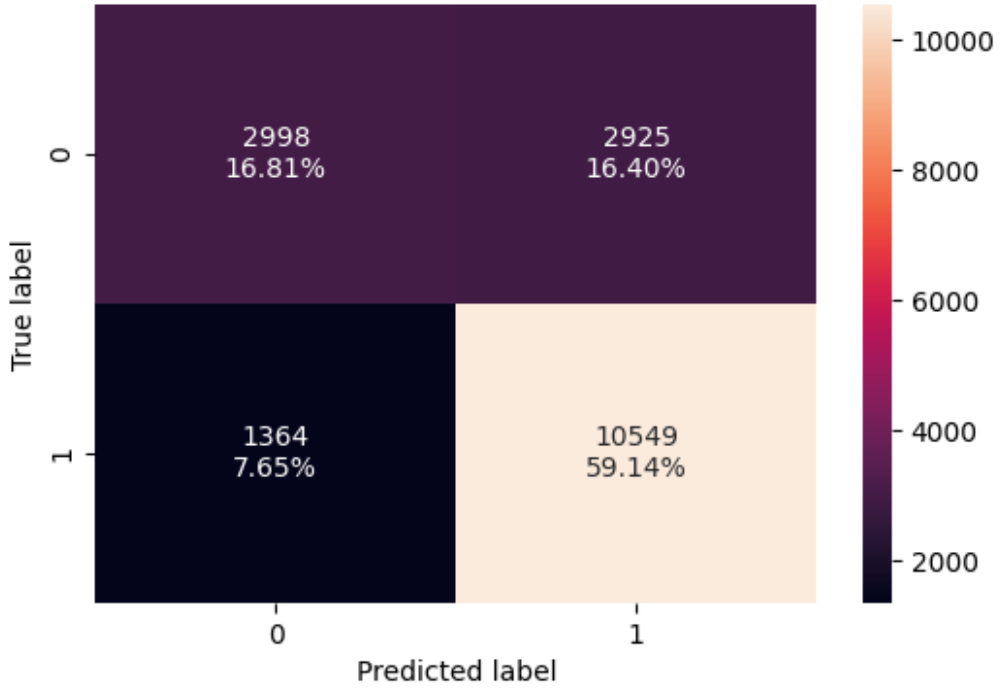

Figure 43(gradient boost tuned train)

accuracy	Recall	Precision	F1
0.76547	0.88164	0.791112	0.83393

Checking model performance on test set

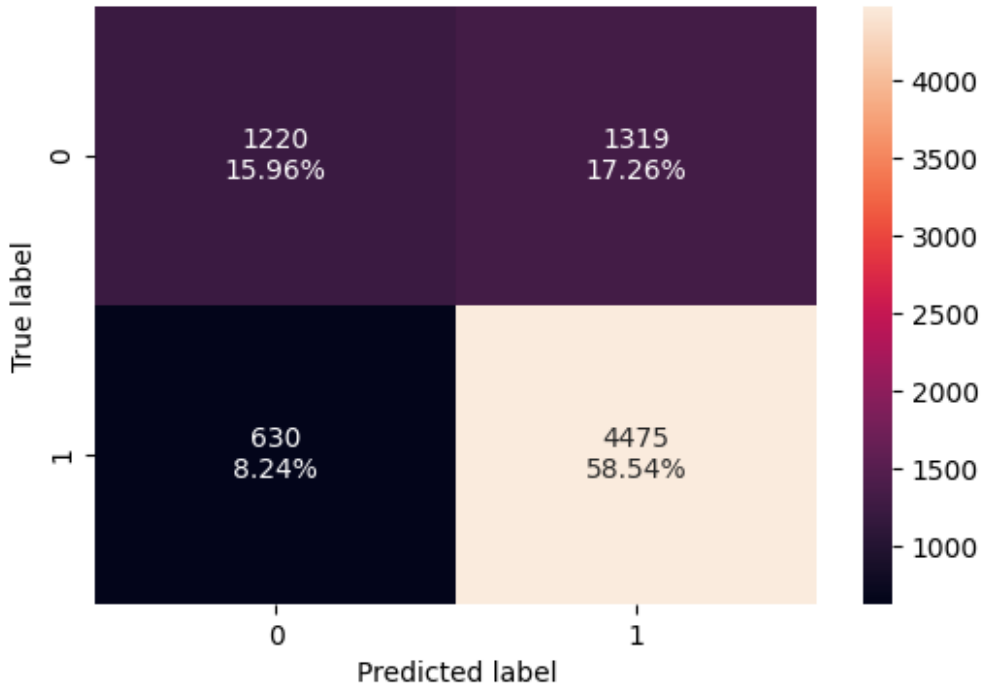

Figure 44(gradient boost tuned test)

accuracy	Recall	Precision	F1
0.74516	0.86954	0.77591	0.820063

- After running the grid search, the tuned xg boost classifier has scale pos weight=1, learning rate=0.1, colsample bytree=0.9, colsample bylevel=1, colsample bynode=1 and random state=1.
- The performance on the test data has definitely increased with F1=0.8200.

Stacking Classifier

The Stacking classifier was defined using Ada boost classifier, tuned gradient boost classifier and tuned random boost classifier as estimators was building the model.

Tuned XGBoost model was considered as the final estimator.

Checking model performance on training set

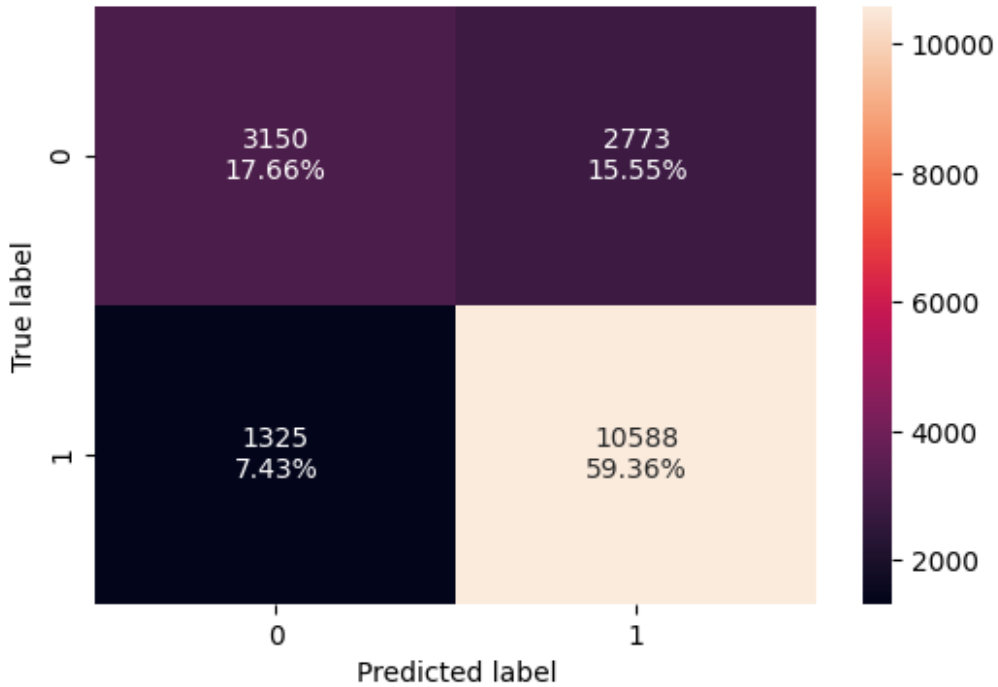

Figure 45(stacking cls train)

accuracy	Recall	Precision	F1
0.77007	0.89414	0.78950	0.838575

Checking model performance on test set

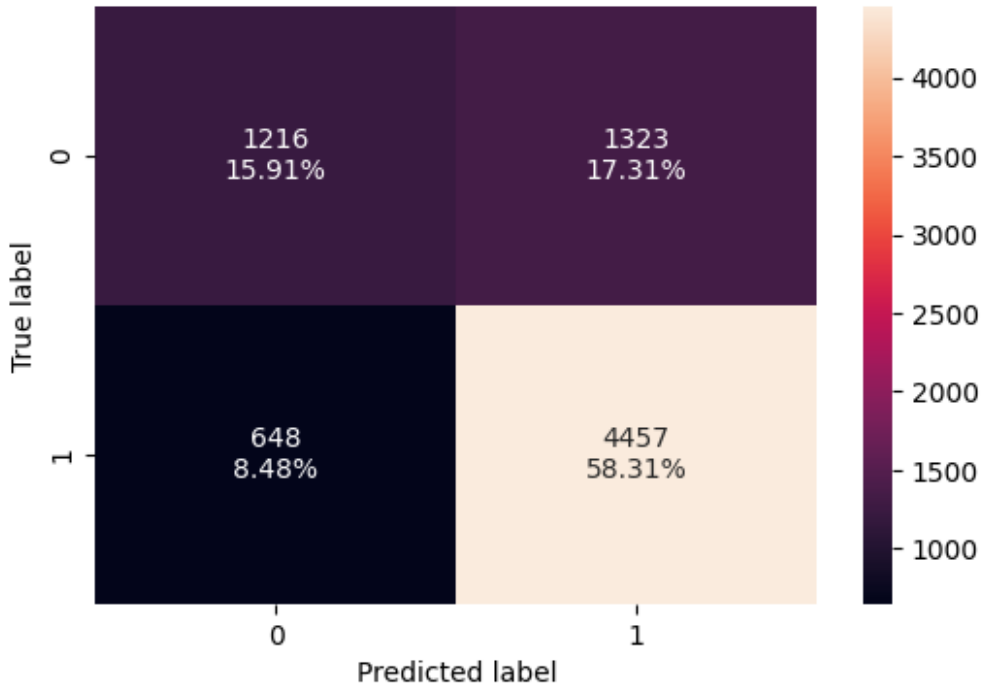

Figure 46(stacking cls test)

accuracy	Recall	Precision	F1
0.74424	0.87992	0.76996	0.82128

- The performance on the train and test data are close and looks like the model is performing good with a F1=0.82128 on the test data.
- The stacking model looks complex with both bagging and boosting classifiers even though the performance is good.
- Let's compare all the models to check which model can satisfy are business goal.

Conclusion

Comparing all models

	Decision Tree	Tuned Decision Tree	Bagging Classifier	Tuned Bagging classifier	Random Forest	Tuned Random forest	Adaboost Classifier	Tuned Adaboost Classifier	Gradient Boost Classifier	Tuned Gradient Boost Classifier	XGBoost Classifier	XGBoost Classifier Tuned	Stacking Classifier
Accuracy	0.706567	0.706567	0.69152	0.72422	0.72082	0.73809	0.734301	0.716641	0.744767	0.743459	0.73325	0.74516	0.74424
Recall	0.930852	0.930852	0.76415	0.89539	0.83212	0.89892	0.885015	0.781587	0.876004	0.871303	0.86072	0.86954	0.87992
Precision	0.715447	0.715447	0.77171	0.74385	0.76886	0.75539	0.757799	0.791510	0.772366	0.773296	0.76791	0.77591	0.76996
F1	0.809058	0.809058	0.76791	0.81262	0.79924	0.82093	0.816481	0.786517	0.820927	0.819379	0.81167	0.82006	0.82128

Figure 47(all model performance)

- The models were evaluated on key metrics—accuracy, precision, recall, and F1-score—with a focus on balancing false positives and false negatives. The AdaBoost classifier displayed acceptable performance but showed noticeable variability on the test set when compared to the gradient boosting approaches. This suggests that while AdaBoost has merit, it might not consistently capture the underlying patterns of the data as robustly as some of the other ensemble methods.

- The tuned Gradient Boosting and Random Forest models both provided solid performance with well-balanced precision and recall scores. Gradient Boosting, in particular, showed a stable trade-off between the two, hinting at its reliability in handling subtle decision boundaries. On the other hand, Random Forest—after hyperparameter tuning—demonstrated robust recall, which is beneficial when minimizing the risk of falsely denying a visa application. These observations indicate that both methods are strong candidates when model consistency and stability are important.

- The tuned XGBoost model, optimized with log loss as the evaluation metric, excelled in delivering high precision. This is particularly advantageous in scenarios where it is critical to reduce the number of false positives, ensuring that potentially suitable applicants are not overlooked. The consistency of XGBoost's performance across the training and test splits makes it an attractive option when precise classification is required.

- The Stacking Classifier, which integrated the strengths of AdaBoost, Gradient Boosting, and Random Forest using XGBoost as the final estimator, emerged with an overall slight edge in achieving balanced performance metrics. Given its ensemble nature, the stacking approach can be recommended when computational resources and integration complexity are manageable, as it leverages the complementary strengths of multiple algorithms. In environments requiring faster deployment or easier interpretability, one might consider using the tuned XGBoost or Random Forest models as viable alternatives.

Important features of the final model

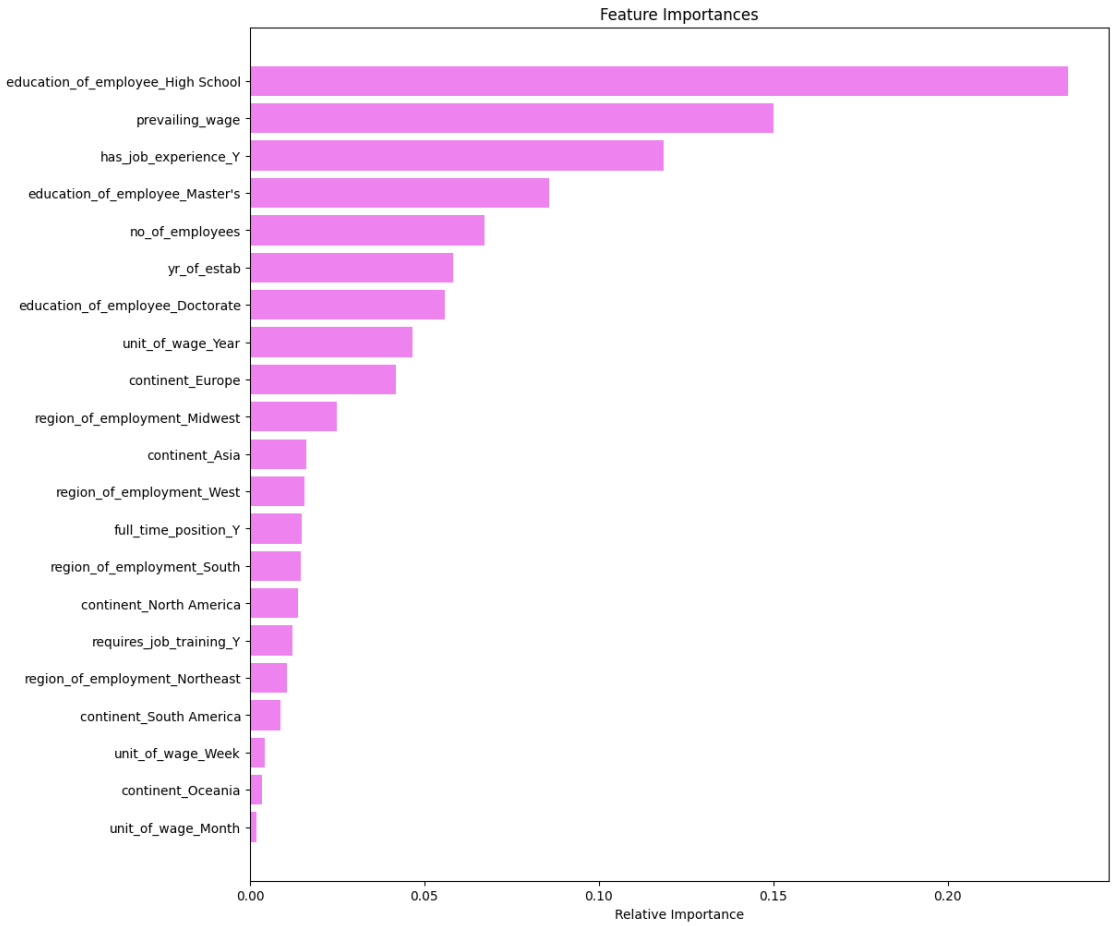

Figure 48(feature importance)

Important features of the final model

Education of the employee – high school, is the most important feature followed by prevailing wage, has job experience-Y and education of the employee- masters are also some of the important features.

Recommendations

- **Adopt an Ensemble Approach:** The experiments indicate that ensemble techniques—especially stacking methods that combine Gradient Boosting, Random Forest, and XGBoost—tend to yield more balanced performance metrics. This layered approach can help reduce both false positives and false negatives, thereby accelerating the visa screening process with a higher confidence level. It is recommended that EasyVisa integrates these ensemble models into its decision-

support system, ensuring that the predictive tool leverages diverse signals from the data.

- **Enhance Model Transparency and Interpretability:** Given the sensitive nature of visa approvals and their impact on both the workforce and applicants, it is essential to embed clear explanation mechanisms within the model. Providing interpretable results, such as key drivers behind a prediction, can aid immigration experts and HR personnel in understanding and validating model decisions. This transparency is fundamental not only for internal trust but also for compliance with regulatory standards, making it a key element of the proposed solution.
- **Implement Continuous Monitoring and Data Refresh Cycles:** As the applicant profiles and labor market trends evolve, periodic model retraining will be crucial in maintaining predictive accuracy. Establishing a routine for data refreshes and model performance evaluation ensures that the system adapts to new trends and minimizes prediction drift. Continuous monitoring can also facilitate timely adjustments in threshold values to better balance the risks associated with certification errors.
- **Include Business Impact Analysis and Risk Mitigation Strategies:** Considering the heavy implications of false predictions—where certifying an unsuitable candidate or rejecting a qualified one can directly affect economic competitiveness—a cost-sensitive approach is advised. By conducting regular business reviews that analyze the financial and operational impacts of model misclassifications, EasyVisa can fine-tune decision thresholds and integrate expert feedback. This proactive approach not only mitigates risks but also aligns the predictive model with broader business goals, ensuring that technological advancements translate into tangible operational improvements.